TRIPMATE AI

ELEC5620 Model Based Software Engineering

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Individual Contributions

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Author-name key for the supplied diagrams: Moore = Mo Shen; Jason = Jason Yang; Runda / rundaweng = Runda Weng.

1 Application introduction

TripMate AI is an educational travel-service prototype that uses four collaborating AI agents to support trip planning, itinerary revision, disruption recovery, complaint handling, membership management, and supplier cooperation. Travellers communicate through the Operations Agent, while specialised agents complete planning, exception handling, and supplier-related tasks. Deterministic tools and a shared Reviewer/Guardrail check important constraints, evidence, prices, and policies. The Human Founder/Operator retains authority over financial, contractual, legal, safety, and other high-impact decisions. The prototype uses simulated travel and supplier data and does not execute real bookings, payments, refunds, contracts, or advertising expenditure.

2 Primary users and roles

2.1 Travellers

Travellers use the controlled Operations interface to submit and confirm travel-planning, plan-revision, disruption, complaint, membership, and routine-enquiry requests. They provide the information and constraints needed for processing, review itinerary recommendations and comparisons, and explicitly confirm any change that would alter a stored preference or hard constraint.

Free Members and Paid Members receive access according to OPS-03 and UC-07. Travellers do not interact directly with specialist Agents; Operations coordinates specialist work, approvals, and external responses.

2.2 Operations Agent

The Operations Agent is TripMate AI's controlled external communication gateway. It receives requests from customers and Travel Service Providers. It also manages routine external communication for the other Agents.

- It uses an LLM to identify request types, extract known information, and find missing fields.
- It asks focused questions when information is missing. It creates a CustomerCase or SupplierCase after the requester confirms the information.
- It receives source-tagged Campaign Responses from approved simulated campaigns, records leads, and reports acquisition metrics. Supplier-data-based campaign proposal design and approval are handled through UC-08.
- It calls the Entitlement Service before a trip is created or extended. The service applies the membership rules defined in OPS-03.
- It routes planning requests to the Travel Planning Agent, disruptions and complaints to the Customer Exception Agent, and supplier requests to the Supplier Partnership Agent.
- It handles routine enquiries with approved service information. It continues to coordinate and track each case after routing.
- Specialised Agents return their findings and proposed responses to the Operations Agent. The Operations Agent checks the evidence, consistency, authority, and wording before release.
- It requests Human Assistance when information is unavailable, classification remains uncertain, the case is high risk, or the requester asks for a human twice. Restricted business decisions require Human Approval.
- It records confirmations, tool results, routing decisions, hand-offs, approvals, and external responses in the Agent Trace.
- It cannot override deterministic services or make unauthorised financial and contractual commitments. Advertising publication and real campaign spending require approval from the Human Founder/Operator.

2.3 Travel Planning Agent

- Acts as TripMate AI's end-to-end travel planner, responsible for turning a validated customer request into complete and feasible itinerary options.
- Retrieves timestamped transport, accommodation, and activity offers from approved simulated supplier data, then combines eligible items into complete door-to-door itineraries.
- Applies every hard constraint before ranking, including dates, budget ceiling, baggage, accessibility, arrival time, transfer limit, availability, and cancellation status; then scores valid options in Comfort, Value, or Budget mode.
- Produces a recommended plan and eligible alternatives with cost breakdowns, sources, timestamps, trade-offs, and exclusion reasons; customer revisions create a new version without silently changing confirmed requirements.
- Maintains a versioned TravelPlan record that links confirmed Traveller preferences, retrieved offers, validation results, ranked itinerary options, comparison criteria, and the explanation returned to the Operations Agent.
- Uses LLM reasoning to interpret preferences, assemble candidate plans, and draft explanations. Deterministic services remain the source of truth for feasibility, price totals, availability, policy checks, and version comparison.
- Escalates through the Operations Agent when no feasible itinerary exists, required data is stale or missing, constraints conflict, confidence remains low after rechecking, or a financial, contractual, safety, or policy-sensitive decision is required.

2.4 Customer Exception Agent

The Customer Exception Agent is responsible for handling travel disruptions and investigating customer complaints within TripMate AI. Its core responsibilities are disruption recovery (EXC-01) and complaint investigation and resolution recommendation (EXC-02). These responsibilities correspond to UC-03, *Recheck an Active Trip and Recover from Disruption*, and UC-04, *Investigate a Complaint and Recommend Resolution*. The agent supports Travellers by coordinating recovery proposals and complaint-resolution recommendations under the supervision of the Human Founder/Operator.

For disruption recovery, the agent reviews the affected itinerary, gathers relevant evidence, and coordinates a recovery proposal. It works with the Travel Planning Agent on itinerary revisions and with the Supplier Partnership Agent to obtain information on alternative services, availability, prices, and supplier terms. For complaints, it investigates the reported issue, collects supporting evidence, and prepares a proposed resolution or refers the case for human handling.

The Customer Exception Agent receives cases routed by the Operations Agent and returns its findings, evidence, unresolved issues, and proposed responses. The Operations Agent remains responsible for communication with Travellers. Recovery and resolution proposals are subject to the shared Reviewer/Guardrail checks, including relevant constraints, evidence, calculations, and policies.

The agent operates within the project's human-approval boundaries. Payments, refunds, compensation, and other significant financial or contractual commitments require human approval. Matters involving legal or safety concerns, or unresolved low confidence, are referred to the Human Founder/Operator. Relevant evidence, tool results, recommendations, approval decisions, and outcomes are recorded to maintain traceability and support human oversight.

2.5 Supplier Partnership Agent

- Assesses supplier offers received through the Operations Agent and prepares approved offers for internal travel planning.
- Uses an LLM to extract supplier information, identify missing or ambiguous details, and explain findings with supporting evidence.
- Uses deterministic functions to validate offer details and calculate selling prices and margins.
- Submits offers and price breakdowns to the Reviewer/Guardrail for evidence, calculation and policy checks.
- Plans and evaluates marketing campaigns using relevant approved supplier data, including eligible products, availability, prices and commercial terms.
- Uses an LLM to draft campaign content and explain the proposal, while deterministic checks evaluate budget limits and consistency with supplier data.
- Requests Human Founder/Operator approval for campaign proposals, binding agreements, restricted commercial terms and financial commitments. External campaign publication or real spending requires explicit approval.
- Routes supplier clarification requests and proposed external messages through Operations. Campaign Responses are handed to Operations with their campaign and source references for handling under UC-05.
- Records offer outcomes, price breakdowns, campaign evaluations, review results and approval decisions in the Agent Trace.
- Cannot independently enter contracts, spend funds, publish campaigns or override failed validation and pricing rules.

2.6 Human Founder/Operator

The Human Founder/Operator supervises TripMate AI and retains authority over decisions that the Agents must not make independently. The operator reviews or takes over cases involving restricted financial, contractual, legal, safety, publication, spending, refund, compensation, or unresolved low-confidence decisions.

The operator provides campaign briefs for UC-08 and may approve, request revision, or reject reviewed supplier offers, campaign proposals, and other restricted commitments. An approval is scoped to the recorded proposal or decision; it does not automatically authorise a booking, payment, contract, advertisement publication, or real spending action.

Human decisions, reasons, scope, and outcomes are recorded in the Agent Trace. The operator may request revision or assume manual handling when available evidence is insufficient for safe automated processing.

2.7 Reviewer/Guardrail

Reviewer/Guardrail is a shared internal control used by the specialist Agents before a result is represented as reviewed or approved. It checks the applicable evidence, deterministic calculations, hard constraints, policy conditions, content restrictions, and approval requirements.

Reviewer/Guardrail returns a pass or revision reasons. It does not replace deterministic services, change source facts, grant entitlement, approve restricted commitments, or communicate directly with external actors. A failed review returns the work to the responsible Agent or triggers human handling when the issue cannot be resolved. Review inputs, results, and reasons are recorded for traceability.

3 Key capabilities and core features

3.1 For Travellers

- Submit and confirm travel-planning, plan-revision, disruption, complaint, membership, and routine-enquiry requests through the controlled Operations interface.
- Receive complete itinerary recommendations and eligible alternatives with cost breakdowns, evidence, trade-offs, and exclusion reasons.
- Request a revision without overwriting the baseline TravelPlan or silently relaxing a confirmed hard constraint.
- Receive a reviewed recovery proposal or complaint-resolution recommendation, or a documented referral when automated handling cannot safely continue.
- Access trips and Premium Features according to the deterministic membership rules in OPS-03 and UC-07.

3.2 For the Operations Agent

3.2.1 Unified external request intake and coordination feature (OPS-02)

- The Operations Agent receives requests from Travellers and Travel Service Providers.
- A source-tagged Campaign Response from an approved simulated campaign is received through UC-05 and retains its campaign ID and source channel. Campaign proposal design, supplier-offer selection, budget evaluation, review, and approval remain within UC-08.
- It classifies each request as travel planning, plan revision, travel disruption, complaint, membership support, routine enquiry, supplier cooperation, inventory, offer, or price negotiation.
- The LLM extracts known information, missing fields, unresolved questions, and classification confidence.
- Deterministic rules define the required fields for each request type. The Operations Agent asks focused questions until the information is complete.
- The requester must confirm the captured information before the case is sent to a specialised Agent. A confirmed field changes only with the requester's approval.
- If a request contains independent intents, the Operations Agent proposes separate but related cases.
- After confirmation, the system creates or updates a CustomerCase or SupplierCase.
- Planning and revision cases go to the Travel Planning Agent. Disruptions and complaints go to the Customer Exception Agent. Supplier cases go to the Supplier Partnership Agent.
- Each hand-off includes the confirmed request, category, relevant case history, available sources, unresolved questions, and membership result when applicable.
- A hand-off does not close the case. The Operations Agent tracks progress and manages later communication with the requester.
- Specialised Agents return their findings and proposed responses to the Operations Agent. The Operations Agent checks the response against the confirmed request, evidence, business rules, and approval status.
- An incomplete or inconsistent response returns to the responsible Agent for revision. Restricted commitments require Human Approval before release.
- The Operations Agent requests Human Assistance when information is unavailable, classification remains uncertain, automated handling is unsafe, or humanRequestCount >= 2.
- The system records confirmations, tool results, routing decisions, hand-offs, human decisions, external responses, and closure events in the Agent Trace.
- The case closes after the approved result is sent, the requester withdraws the request, or the Human Founder/Operator closes it.

3.2.2 Membership entitlement and premium access feature (OPS-03)

- Before creating or extending a trip, the Operations Agent sends the customer ID, requested action, and trip dates to the deterministic Entitlement Service.
- A free member may hold no more than two simultaneous trips with a Planned or Active status.
- A free-member trip may cover no more than five calendar days.
- Trip duration is calculated as: (end date - start date + 1 calendar day).
- Paid members are not subject to the free-tier trip-count or duration limits while their subscriptions are active.
- Active paid members receive an ad-free interface, route-map access, and other approved premium functions.
- Advertising shown to free members must be clearly labelled. It cannot alter hard-constraint filtering or recommendation ranking.
- The Entitlement Service returns Allowed, Denied with a reason, or VerificationFailed.
- If a free member reaches a limit, the system keeps all existing trips. It explains the failed rule and offers valid actions, such as completing an existing trip, shortening the requested trip, or upgrading.

- If membership status cannot be verified, the restricted action is paused and the service failure is recorded.
- The LLM may explain an entitlement result in plain language. It cannot infer membership status, grant premium access, or override the Entitlement Service.

3.3 For the Travel Planning Agent

3.3.1 Complete itinerary construction, validation, and comparison feature (TP-01)

- Retrieves only eligible transport, accommodation, and activity resources, preserving the supplier source, retrieval time, availability, price, cancellation terms, and reliability information for each item.
- Builds complete itinerary candidates that connect every travel leg, stay, and selected activity, rather than recommending isolated items.
- Uses deterministic validation services to calculate total cost, check schedules, transfer time and opening hours, and reject conflicts or candidates that breach hard constraints.
- Only feasible candidates are ranked in Comfort, Value, or Budget mode. The result states total and component costs, constraint coverage, source freshness, trade-offs, and exclusion reasons for rejected alternatives.
- The Agent explains why the recommended plan is preferred and compares it with eligible alternatives using common criteria.

3.3.2 Controlled plan revision and version comparison feature (TP-02)

TP-02 accepts a Traveller-confirmed change to an existing TravelPlan without modifying the baseline.

TP-02 creates a new immutable version after a Traveller-confirmed revision and provides a field-level comparison with the baseline, including schedule, selected resources, price, constraint coverage, assumptions, and trade-offs.

The Agent creates a revised itinerary version only after the Traveller confirms the changed requirement and never silently relaxes a hard constraint. The revised version must pass the same applicable evidence, price, schedule, and hard-constraint checks before it is presented as feasible.

3.4 For the Customer Exception Agent

3.4.1 Disruption recovery feature (EXC-01)

This feature supports the reassessment of an active trip when a travel disruption affects the existing itinerary. The Customer Exception Agent reviews the relevant itinerary and disruption information, collects supporting evidence, and identifies the travel arrangements that require reassessment.

The agent coordinates with the Travel Planning Agent to prepare revised itinerary options and with the Supplier Partnership Agent to obtain relevant information on alternative services, availability, prices, and supplier terms. Recovery proposals must account for the Traveller's existing preferences and constraints, including budget and timing. The shared validation tools and Reviewer/Guardrail check the proposed options against relevant constraints, evidence, calculations, and policies.

The output is a validated recovery proposal, accompanied by supporting evidence and any unresolved issues or approval requirements. The proposal is returned to the Operations Agent for communication with the Traveller. Where a suitable recovery proposal cannot be established or the case requires a restricted commitment, the agent refers the matter to the Human Founder/Operator. Payments, refunds, compensation, and binding commitments require human approval. Relevant findings, tool results, proposals, and decisions are recorded for traceability.

3.4.2 Complaint investigation and resolution recommendation feature (EXC-02)

This feature supports the investigation of Traveller complaints and the preparation of resolution recommendations for complaints routed by the Operations Agent. The Customer Exception Agent examines the complaint, reviews relevant trip and service information, and collects evidence needed to understand the issue. Where information is incomplete, it requests clarification through the Operations Agent or obtains relevant findings from the responsible specialist Agent.

The agent uses the available evidence and applicable policies or supplier terms to prepare a resolution recommendation. It consults the Travel Planning Agent when the complaint concerns itinerary arrangements and the Supplier Partnership Agent when supplier services, prices, or terms require clarification. Proposed resolutions are subject to the relevant Reviewer/Guardrail checks.

The output is a complaint investigation result and proposed resolution, or a referral for human handling with the available evidence and unresolved issues. Refunds, compensation, and other significant commitments require approval from the Human Founder/Operator. Cases involving legal or safety concerns, or unresolved low confidence, are also escalated. The Operations Agent manages communication of the reviewed outcome to the Traveller, while the investigation evidence, recommendations, and approval decisions are recorded for traceability.

3.5 For the Supplier Partnership Agent

3.5.1 Supplier offer import and validation feature (SUP-01)

- Operations supplies a confirmed SupplierCase and simulated supplier offer source.
- The LLM extracts supplier/product identifiers, availability, supplier price, currency, validity period and cancellation terms, and identifies missing or ambiguous information. Extracted fields remain linked to the source.
- Deterministic checks require the configured fields, an eligible supplier, positive available quantity, a non-negative price, an unexpired validity period and permitted terms. Initial prototype data uses one consistent AUD price basis with all mandatory charges already included. Unknown charges or unsupported currency require correction.
- Failed checks produce RevisionRequired with a reason. Operations coordinates clarification and confirms supplier-supplied corrections before reassessment.
- Only Approved, unexpired offers may be selected by Travel Planning. A material correction clears previous review/approval and repeats assessment. Expiry is checked when the offer is retrieved; automatic notifications and background monitoring are not required for this prototype.
- Save the source, extracted fields, status and key decisions. A tool or save failure prevents approval and is shown for manual retry.

3.5.2 Pricing and margin validation feature (SUP-02)

- A deterministic function calculates basePrice = round(supplierCost x (1 + markupRate), 2), sellingPrice = round(basePrice - discountAmount, 2), and marginAmount = sellingPrice - supplierCost. All amounts use the same AUD and quoted-quantity basis. The selling price must be positive and the configured minimum margin must pass after the discount. The prototype supports one approved fixed-amount discount per quotation, with no discount stacking.
- The configured minimum margin amount must pass after rounding. Missing pricing configuration or a failed margin rule produces RevisionRequired; the LLM cannot change the recorded supplier cost to make the calculation pass.
- PriceBreakdown contains supplierCost, markupRate, basePrice, discountAmount, sellingPrice, marginAmount and currency. A base offer uses discountAmount = 0. When an approved UC-08 member offer applies, the system uses the eligibility result from UC-07 before applying the recorded discount. Missing or failed eligibility verification pauses the member-specific quotation.
- Reviewer/Guardrail checks source consistency, the calculated values and applicable rules. Failure returns the offer for revision.
- Binding supplier prices, agreements, restricted terms and financial commitments require recorded human approval. The human can Approve, Request revision or Reject. Requested revision repeats the checks; it never flows directly to Approved. No decision leaves the offer AwaitingApproval.
- Save the approved outcome and trace before returning the offer to Operations for its normal communication checks. Internal approval is not a booking, payment or signed agreement.

3.5.3 Supplier-data-based marketing campaign planning and evaluation (SUP-03)

- The Human Founder/Operator provides the campaign objective, audience, permitted channel, campaign dates, budget ceiling and content restrictions.
- Supplier Partnership selects relevant Approved, unexpired supplier offers from the simulated dataset. It retains the source references, prices, availability, validity and commercial terms used in the proposal. Unclear information is clarified through Operations.
- The LLM drafts a campaign proposal containing the selected offers, target audience, channel, dates, draft promotional content and an explanation of how the proposal supports the stated objective. It must not invent availability, discounts, supplier commitments or unsupported claims.
- The Agent uses configured simulated channel costs to estimate the campaign budget. Deterministic checks verify the budget ceiling, permitted channel, dates and consistency with supplier prices and terms. A campaign may include a member offer specifying the eligible membership tiers, applicable supplier offers, fixed discount amount and validity period. The rule requires human approval. UC-07 evaluates member eligibility. UC-06 calculates a separate customer-specific PriceBreakdown using the approved MemberOfferRule and UC-07 eligibility decision. It rechecks the final selling price and minimum margin without overwriting the reusable SupplierOffer base price. A failed discounted quotation does not invalidate an otherwise valid approved SupplierOffer.
- Reviewer/Guardrail checks the supplier evidence, calculations, advertising claims and configured content/privacy restrictions. A failed check returns the proposal for revision; a passing proposal goes to the Human Founder/Operator for approval.
- The human can Approve, Request revision or Reject. A material change to campaign content, offers, dates, channel or budget clears the previous approval and repeats evaluation and review. No human response is treated as approval.

- Save the proposal, cost breakdown, evaluation, source references, review and human decision in the Agent Trace. Approval of the proposal does not itself publish an advertisement or spend money. The prototype may preview or simulate an approved campaign; any external publication or real spending needs explicit authorisation for that action.
- Any simulated Campaign Response carries its campaign ID and source channel to Operations for UC-05 intake and coordination. Operations does not take ownership of campaign planning, budget evaluation or approval.

3.6 For the Human Founder/Operator

- Review, approve, request revision, reject, or take over restricted proposals and Cases.
- Provide campaign briefs and decide reviewed UC-08 campaign proposals.
- Retain control of real publication, spending, contracts, payments, refunds, compensation, and other high-impact commitments.
- Receive evidence, unresolved issues, review results, and escalation reasons before making a decision.
- Record the decision, reason, scope, and outcome in the Agent Trace.

4 Optional features

4.1 For the Operations Agent

- **Demand, acquisition, and retention analysis (OPS-O1).** The Operations Agent may analyse simulated campaign, lead, membership, and trip records for human review. An approved simulated campaign may create source-tagged leads; each lead records its campaign, source channel, creation time, consent status, and qualification status. Operations may record impressions, clicks, total leads, qualified leads, converted customers, and simulated campaign cost.

The derived acquisition measures are:

- Click-through rate = recorded clicks / recorded impressions × 100%.
- Qualified-lead rate = leads meeting the recorded qualification rule / source-tagged leads × 100%.
- Qualified-lead conversion rate = qualified leads becoming customers / leads meeting the recorded qualification rule × 100%.
- Simulated customer acquisition cost = total simulated campaign cost / qualified leads becoming customers.

For retention analysis, let A_0 be customers active at the start of the stated reporting period and A_1 be members of the same cohort who remain active at the end. Retention rate = A_1 / A_0 × 100%. Customers acquired during the period are not included in A_1. A zero denominator produces a not-applicable result. Each result states the reporting period, included records, cohort definition, assumptions, and missing data. The LLM may explain observed patterns and recommend continuing, pausing, or revising a future campaign, but it cannot change recorded values, present correlation as causation, publish a campaign, or increase real spending.

- **OPS-O2: Campaign experiment analysis recommendations.** The Operations Agent may analyse recorded campaign and lead results and recommend a hypothesis, control condition, one changed variable, target segment, success metric, duration, minimum data requirement, and stopping condition as input to a future UC-08 campaign brief. The Operations Agent does not select SupplierOffers, create or approve the campaign proposal, publish the experiment, or spend funds. Supplier-data-based proposal design, budget evaluation, review, and approval remain within UC-08.

4.2 For the Travel Planning Agent

- **Destination and travel-policy guidance (TP-O1).** A retrieval-augmented knowledge service for destination guidance and travel-policy information, with cited source material and a clear distinction between general guidance and live availability.
- **Live provider integration (TP-O2).** Integration with real transport, accommodation, activity, weather, and map-data providers beyond the controlled prototype dataset, subject to data freshness checks, supplier agreements, privacy controls, and human approval boundaries.
- **Preference learning and explanation (TP-O3).** This feature may summarise repeated Traveller choices and produce comparison narratives but must obtain confirmation before changing a stored preference or weakening a hard constraint.
- **Live-data forecasting (TP-O4).** Live-data and forecasting services must record provider, retrieval time, freshness status, uncertainty, and fallback behaviour. Stale, unavailable, or conflicting data must not be presented as confirmed itinerary availability.

4.3 For the Supplier Partnership Agent

- **Supplier reliability summary (SUP-O1).** The Agent may summarise simulated completed supplier outcomes for human review. A deterministic function counts eligible final outcomes and supplier cancellations for a stated supplier and reporting period. Supplier-cancellation rate

= supplier cancellations / eligible final outcomes x 100%. Zero outcomes produce Not applicable. The LLM explains the recorded counts and limitations without fabricating evidence or assigning an unsupported reliability rating. Reviewer checks the result before display. The feature does not change contracts, supplier eligibility or approved offers.

This is a small scorecard implementation: recorded counts plus one rate and an evidence-based explanation. Weighted scores, predictions and automatic supplier suspension are not required.

4.4 For the Customer Exception Agent

- **Proactive disruption notification (EXC-O1).** This optional feature allows TripMate AI to identify relevant disruption information and notify affected Travellers before they submit a support request. It extends the core disruption recovery feature by providing earlier awareness of potential problems affecting an active itinerary.

The Customer Exception Agent checks available travel or weather updates against the Traveller's itinerary, including travel dates, destinations, and scheduled services. For example, a transport cancellation update may relate to a planned trip leg, while a weather warning may affect a scheduled outdoor activity. The agent reviews the source and relevance of the information and gathers supporting evidence before preparing a notification. Confirmed disruptions and potential risks are clearly distinguished.

Each proposed notification explains the affected itinerary item, the reported disruption or risk, its possible impact, and the source of the information. It also identifies any uncertainty and indicates whether further investigation or recovery planning is needed. The notification is checked through the shared Reviewer/Guardrail process and passed to the Operations Agent for communication with the Traveller.

If an itinerary requires reassessment, the Customer Exception Agent coordinates further handling through EXC-01, the disruption recovery feature. A notification does not itself authorise changes to bookings, payments, refunds, or compensation. Matters requiring restricted commitments or human judgement are referred to the Human Founder/Operator under the shared approval rules.

The system records the supporting update, affected itinerary, review outcome, and notification delivery result for traceability. For the prototype, simulated travel and weather updates can demonstrate this feature without requiring continuous access to live external services.

5 Feature Diagram

This feature model summarises TripMate AI's mandatory capabilities, optional extensions, and constrained choices. It links Operations, membership, supplier partnerships, travel planning, exception handling, human oversight, and traceability to the feature and use-case identifiers defined in the report.

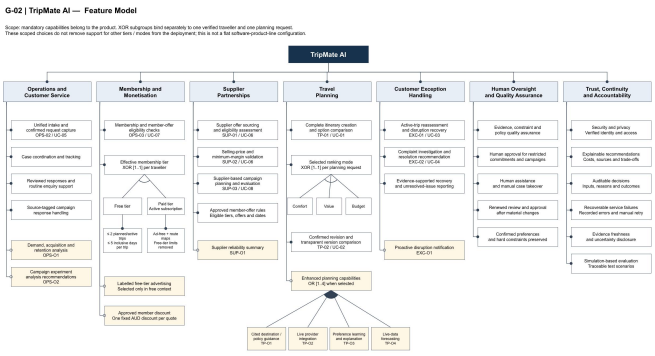


Fig. 1. G-02 — TripMate AI Feature Model

See page 32 for an enlarged version.

6 Ad-hoc requirements

6.1 For the Operations Agent

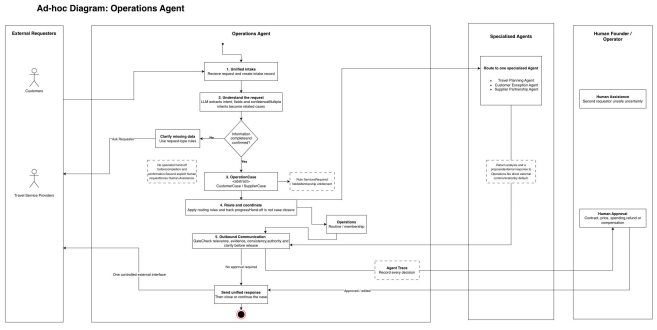


Fig. 2. OPS-AH-01 — Operations Agent Ad-hoc Coordination Flow

See page 33 for an enlarged version.

Requirement ID: OPS-AH-01
Owner / Author: Jiaheng Guo
Related core features: OPS-02, OPS-03
Related use cases: UC-05, UC-07
Related diagrams: OPS-Ad_Hoc, AD-01, SD-01, SM-01, G-03

The Operations Agent should act as the controlled coordination point for requests from Travellers and Travel Service Providers. It should create an intake record and identify the request's intent, available information, missing required fields,

and classification confidence. It should ask focused questions when information is missing and must not route an incomplete or unconfirmed request.

If a request contains independent intents, the Operations Agent should propose separate but related CustomerCase or SupplierCase records. The records should be created only after the requester confirms the split and the captured information. A second explicit request for human support should transfer the Case to Human Assistance with the information collected so far.

After confirmation, the Operations Agent should apply the configured routing and entitlement rules. Planning and revision Cases should be routed to the Travel Planning Agent; disruptions and complaints should be routed to the Customer Exception Agent; supplier-related Cases should be routed to the Supplier Partnership Agent; and membership or premium-access checks should be handled through UC-07. A routine enquiry that can be answered from Approved Service Information does not require specialist routing.

The Operations Agent should continue tracking each Case after hand-off. A specialised Agent should return its findings, evidence, unresolved issues, and proposed response to Operations. Before external release, Operations should check the response for relevance to the confirmed request, supporting evidence, consistency with deterministic rules, required authority, approval status, and clarity. Contracts, binding prices, real spending, refunds, compensation, and other restricted commitments require a recorded Human Founder/Operator decision before communication or execution.

The Operations Agent should record confirmations, classifications, tool results, routing decisions, hand-offs, approvals, external responses, and closure events in the Agent Trace. All external messages should pass through the controlled external interface. A Case should close only after an approved result is delivered, the requester withdraws the request, or the Human Founder/Operator records a closure decision.

6.2 For the Travel Planning Agent

Requirement ID: TP-AH-01
Owner / Author: Mo Shen
Related core features: TP-01, TP-02
Related use cases: UC-01, UC-02
Related diagrams: AD-02, SD-02, SM-02, G-03

Travel-plan construction, constraint validation, comparison, and controlled revision

The Travel Planning Agent shall accept only a confirmed planning or revision case, construct complete door-to-door candidates from approved supplier data, and preserve source, offer identifier, retrieval time, availability, price, cancellation terms, and reliability evidence for every resource considered.

It shall apply deterministic validation before ranking. Validation shall check dates, origin and destination, budget ceiling, party size, baggage, accessibility, arrival-time requirement, transfer limit, availability, opening hours, cancellation status, and policy restrictions; it shall calculate component-level and total price and reject infeasible or unsupported candidates.

For every feasible plan, it shall provide recommendation rationale, comparison criteria, trade-offs, cost breakdown, sources, freshness evidence, and reasons for rejected candidates. Preference modes may affect ranking only after all hard constraints pass.

A Traveller-confirmed change shall create a new immutable TravelPlan version. The baseline shall be preserved and the comparison shall show changes to schedule, resources, price, constraint coverage, assumptions, and trade-offs; confirmed hard constraints shall never be silently relaxed.

If no feasible plan exists or data, evidence, constraints, or confidence remain unresolved, the agent shall return an actionable issue report to the Operations Agent. Financial, contractual, safety, and policy-sensitive decisions require Human Founder/Operator approval, and all inputs, validation outcomes, versions, comparisons, escalations, approvals, and results shall be recorded in the Agent Trace.

6.3 For the Customer Exception Agent

Requirement ID: EXC-AH-01
Owner / Author: Runda Weng
Related core feature: EXC-01
Related use case: UC-03
Related diagrams: AD-03, SD-03, SM-03, G-03

Active-trip disruption recovery

TripMate AI should help Travellers manage disruptions that affect an active trip. When an existing travel arrangement becomes unavailable or can no longer proceed as planned, the Traveller needs a recovery proposal that considers the affected itinerary and their original travel constraints. The Customer Exception Agent should investigate the disruption, gather relevant evidence, and coordinate recovery options with the other specialist agents.

The agent should receive the confirmed disruption-related request and its available context from the Operations Agent. It should review the current itinerary and the information supporting the reported disruption to identify which arrangements are affected. For example, a cancelled transport service may affect the timing of accommodation or activities later in the trip. Where additional Traveller information is needed, the agent should request clarification through the Operations Agent.

Recovery planning should involve the appropriate specialist agents. The Supplier Partnership Agent should provide relevant alternative services, availability, prices, and supplier terms, while the Travel Planning Agent should support the preparation of revised itinerary options. The Customer Exception Agent should coordinate these contributions and assemble a recovery proposal that explains the proposed changes

and their implications for the Traveller's budget, timing, and preferences.

A proposed recovery option should undergo the relevant deterministic validation and shared Reviewer/Guardrail checks before being presented to the Traveller. These checks should address applicable constraints, evidence, availability, calculations, and policies. A proposal that does not pass the required checks should be revised or referred for human handling. If the available information or resources are insufficient to establish a suitable option, the agent should identify the unresolved issues and provide the available evidence when escalating the case.

The Customer Exception Agent should operate within the project's human-approval rules. It should not independently authorise payments, refunds, compensation, or binding supplier commitments. Decisions involving significant financial, contractual, legal, or safety implications, as well as unresolved low-confidence cases, should be referred to the Human Founder/Operator. Any required approval should be recorded before the corresponding commitment is communicated or acted upon.

The agent should return its findings, supporting evidence, proposed recovery options, and outstanding approval requirements to the Operations Agent, which manages communication with the Traveller. The system should maintain a traceable record connecting the original request, relevant evidence, tool results, itinerary versions, review outcomes, human approvals, and final handling outcome. The expected result is a validated recovery proposal or a documented escalation for human assistance. The teaching prototype should demonstrate this behaviour using simulated supplier, booking, payment, and disruption data.

Requirement ID: EXC-AH-02
Owner / Author: Runda Weng
Related core feature: EXC-02
Related use case: UC-04
Related diagrams: AD-03, SD-03, SM-03, G-03

Complaint investigation and resolution recommendation

TripMate AI should help a Traveller obtain an evidence-supported response to a confirmed complaint. The Customer Exception Agent should receive the confirmed complaint and available Traveller context from the Operations Agent, identify the issue requiring investigation, and gather the relevant itinerary information, service records, Traveller-provided material, policies, and supplier terms.

When itinerary arrangements require clarification, the Customer Exception Agent should obtain findings from the Travel Planning Agent. When supplier services, prices, or terms require clarification, it should obtain findings from the Supplier Partnership Agent. Missing information should be requested through Operations, and conflicting or inconclusive evidence should be recorded rather than presented as a supported conclusion.

The Customer Exception Agent should prepare either an evidence-supported resolution recommendation or a documented referral for human handling. The Reviewer/Guardrail should check the evidence, applicable rules, calculations, and approval requirements. Refunds, compensation, binding commercial commitments, and decisions involving legal, safety, or unresolved low-confidence concerns require the Human Founder/Operator. The Operations Agent remains responsible for approval coordination and communication with the Traveller.

The complaint, evidence, specialist findings, recommendation or referral, review outcome, approvals, and hand-offs should remain connected in the Agent Trace. This requirement does not authorise the Customer Exception Agent to execute a refund, compensation payment, booking change, or binding supplier commitment.

6.4 For the Supplier Partnership Agent

Requirement ID: SUP-AH-01
Owner / Author: Jason Yang
Related core features: SUP-01, SUP-02
Related use case: UC-06
Related diagrams: AD-04, SD-04, SM-04, G-03

Supplier offer assessment and margin validation

The Supplier Partnership Agent should assess supplier offers received through Operations and prepare reliable offers for internal planning. It should use an LLM to extract the supplier, product, availability, price, validity and cancellation details while retaining the source. Deterministic checks should identify missing, ambiguous, expired or otherwise invalid information. The Agent should request correction through Operations and keep the offer unavailable until the corrected information has been confirmed and checked.

For a valid offer, the system should calculate the selling price and margin using configured rules and create a Price-Breakdown. Reviewer/Guardrail should check the evidence, arithmetic and policy conditions. Restricted decisions should wait for Human Founder/Operator approval. A requested revision should repeat assessment; rejection should be recorded with a reason. Only an approved, unexpired offer should be available to Travel Planning. The system should retain a simple trace and return outcomes through Operations. The prototype uses simulated data and does not execute bookings, contracts or payments.

Requirement ID: SUP-AH-02
Owner / Author: Jason Yang
Related core feature: SUP-03
Related use case: UC-08
Related diagrams:

G-03 (overall use-case coverage). AD-04, SD-04, SM-04 and G-05A/B/C describe the UC-06 supplier-offer assessment dependency; they do not directly model the UC-08 campaign approval workflow.

Supplier-data-based campaign planning and evaluation

The Supplier Partnership Agent should use approved and unexpired supplier offers to prepare a marketing campaign proposal based on a Human Founder/Operator campaign brief. The proposal should include selected offers, target audience, channel, dates, promotional content and estimated cost. Deterministic checks should verify the budget, campaign dates, supplier prices and terms. The Reviewer/Guardrail should check supporting evidence and configured advertising restrictions. The Human Founder/Operator should approve, request revision or reject the proposal. Material revisions should repeat the relevant checks and require fresh approval. Where a campaign includes a member offer, the proposal should record its eligibility conditions, discount and validity period for human approval. UC-07 should evaluate member eligibility. UC-06 should calculate a separate customer-specific PriceBreakdown using the approved MemberOfferRule and the UC-07 eligibility decision. It should recheck the final selling price and minimum margin without overwriting the reusable SupplierOffer base price. A failed discounted quotation should not invalidate an otherwise valid approved SupplierOffer. The system should record the proposal, evaluation and decision in the Agent Trace. External publication and real spending remain outside the base prototype.

7 Use case specifications

7.1 Overall Use Case diagram

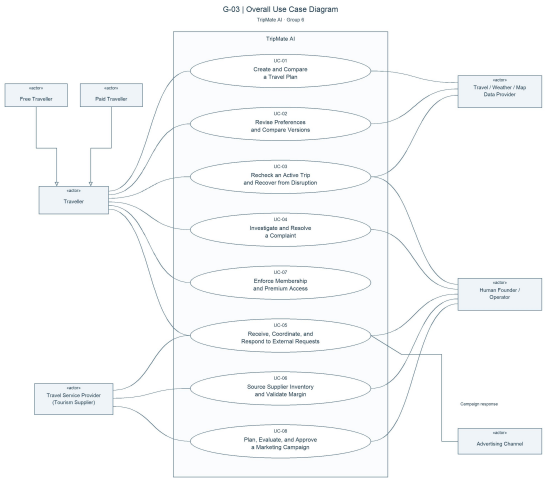


Fig. 3. G-03 — Overall TripMate AI Use Case Diagram

See page 34 for an enlarged version.

This use case diagram defines the TripMate AI system boundary and connects Traveller roles, Travel Service Providers, data providers, the Human Founder/Operator, and the advertising channel to UC-01 through UC-08. It provides coverage and actor context rather than detailed internal behaviour.

7.2 For the Operations Agent

7.2.1 UC-05: Receive, Coordinate, and Respond to External Requests

Owner / Author: Jiaheng Guo
Related requirements: OPS-02, OPS-AH-01
Related diagrams: AD-01, SD-01, SM-01, G-03

Name: Receive, Coordinate, and Respond to External Requests

Goal: Receive an external request, confirm the required information, coordinate the responsible handler, and deliver an approved and traceable response.

Level: User-goal level

Primary Actors: Traveller; Travel Service Provider

Supporting External Actor: Human Founder/Operator

Internal Participants: Operations Agent; Travel Planning Agent; Customer Exception Agent; Supplier Partnership Agent; Entitlement Service; Reviewer/Guardrail; Authentication and Access Control mechanism; Agent Trace

Preconditions:

1. The controlled external interface is available.
2. Request categories, required fields, routing rules, and approval rules are configured.
3. The Operations Agent can access the Case Repository, Approved Service Information, Entitlement Service, and Agent Trace.
4. Authentication and access-control services are available.

Success Condition: The approved response is delivered through the Operations Agent, the processing and delivery records are stored in the Agent Trace, and the Case is closed.

Failure Condition: If the request cannot be verified, confirmed, coordinated, approved, or delivered, the Case remains open in the relevant waiting or escalation state.

Trigger: A Traveller or Travel Service Provider submits a new request or follow-up message through the controlled external interface. The request may be a source-tagged Campaign Response.

Main Success Scenario

1. The requester submits an external request.
2. The Operations Agent identifies the requester type and request source.
3. The system creates an ExternalRequest intake record containing the requester type and request source.
4. The requester completes the required identity and access checks.
5. The Operations Agent uses the LLM to identify the request type and extract the available information.

6. Deterministic rules identify the required fields for the request type.
7. The Operations Agent obtains any missing information from the requester.
8. The requester confirms the structured request and any proposed intent split.
9. The system creates one or more related CustomerCase or SupplierCase records from the confirmed ExternalRequest.
10. Routing Rules assign the Case to the responsible specialist Agent or the Entitlement Service.
11. The Operations Agent records the hand-off and continues tracking the Case.
12. The responsible specialist returns its findings, evidence, and proposed response.
13. The Operations Agent checks the response against the confirmed request, available evidence, business rules, and approval status.
14. Human Approval is obtained when the response contains a restricted commitment.
15. The Operations Agent sends the approved response through the controlled external interface.
16. The system records the delivery and closes the Case.

Alternative and Error Flows

A1. Routine enquiry (at Steps 5-10)

If the request can be answered using Approved Service Information, the Operations Agent prepares and sends the response without specialist coordination.

A2. Identity cannot be verified (at Step 4)

The ExternalRequest remains pending and no CustomerCase is created. Protected case, membership, or supplier information is not accessed or processed until identity verification is successfully completed. The user may retry verification or cancel the request.

A3. Required information is missing (at Steps 5-7)

The Operations Agent asks focused questions and continues intake after the requester supplies the missing information.

A4. Classification confidence is low (at Step 5)

The Operations Agent asks the requester to clarify the intent. If the intent remains unclear, the Case is sent to Human Assistance.

A5. The request contains multiple independent intents (at Steps 5-8)

The Operations Agent proposes separate but related Cases. The Cases are created after the requester confirms the split.

A6. Confirmed information changes (after Step 8)

The Operations Agent updates the affected fields and obtains confirmation. If the change affects category, routing, risk, or approval conditions, the requester confirms the complete request summary again.

A7. The specialist requires more information (at Step 12)

The Case enters Awaiting Requester. The Operations Agent obtains and confirms the additional information before returning it to the specialist.

A8. The specialist is unavailable (at Steps 10-12)

The Case remains open and is sent to Human Assistance with the request details and recorded coordination failure.

A9. The proposed response fails review (at Step 13)

The Operations Agent returns the response to the responsible specialist with the revision reason.

A10. Human Approval is required (at Step 14)

The Case enters Awaiting Human Approval. The Human Founder/Operator may approve, edit, reject, or take over the response.

A11. Delivery fails (at Steps 15-16)

The Case remains Ready to Respond while delivery is retried. Persistent delivery failure is sent to Human Assistance.

A12. The requester withdraws the request (at any step before closure)

The withdrawal is recorded and the Case is closed as Requester Withdrawn.

Business Rules

1. The Operations Agent is TripMate AI's controlled external communication gateway.
2. Protected processing requires verified identity and access.
3. Required information must be complete and confirmed before specialist coordination begins.
4. The LLM may classify requests, extract information, identify missing fields, and explain results.
5. Deterministic rules define required fields, membership entitlements, routing conditions, and approval requirements.
6. Planning and revision requests are routed to the Travel Planning Agent.
7. Disruptions and complaints are routed to the Customer Exception Agent.
8. Supplier cooperation, inventory, offer, and price-related requests are routed to the Supplier Partnership Agent.
9. Membership checks are handled through the Entitlement Service.
10. Specialist responses are reviewed by the Operations Agent before delivery.
11. Restricted financial, contractual, legal, access, refund, or compensation decisions require Human Approval.
12. If humanRequestCount >= 2, the Case is sent to Human Assistance.
13. Confirmations, classifications, tool results, routing decisions, hand-offs, approvals, responses, and closure events are recorded in the Agent Trace.

Tools and Data

- Controlled External Interface
- LLM Classification Output
- Authentication and Access Control mechanism
- Required Field Rules
- Routing Rules
- CustomerCase and SupplierCase Repository
- Approved Service Information
- Entitlement Service
- Specialist Findings and Proposed Responses
- Human Assistance Queue
- Human Approval Record
- Agent Trace

Acceptance Criteria

AC-UC05-01 — Request intake

Given an external request, when the Operations Agent begins intake, then an ExternalRequest intake record is created with the requester type and request source.

AC-UC05-02 — Information confirmation

Given a complete ExternalRequest, when the requester confirms the captured information and any proposed intent split, then one or more related CustomerCase or SupplierCase records are created and become eligible for routing.

AC-UC05-03 — Routing

Given a confirmed request, when Routing Rules are applied, then the Case is assigned to the responsible specialist Agent or Entitlement Service and the hand-off is recorded.

AC-UC05-04 — Human assistance

Given unresolved classification uncertainty or humanRequestCount >= 2, when the escalation rule is met, then the Case is sent to Human Assistance with its available information and escalation reason.

AC-UC05-05 — Response control

Given a proposed specialist response, when the Operations Agent reviews it, then only a response that satisfies the confirmed request, evidence, business rules, and approval conditions proceeds to delivery.

AC-UC05-06 — Completion

Given an approved response, when delivery succeeds, then the response and delivery result are recorded and the Case is closed.

7.2.2 UC-07: Enforce Membership and Premium Access

Owner / Author: Jiaheng Guo
Related requirements: OPS-03, OPS-AH-01
Related diagrams: AD-01, SD-01, SM-01, G-03

Name: Enforce Membership and Premium Access

Goal: Use verified membership status, trip facts, and feature entitlements to return an explained and traceable access decision for a protected TripMate AI action.

Level: Subfunction level
Primary Actor: Traveller
Supporting External Actor: Human Founder/Operator
Internal Participants: Operations Agent; Travel Planning Agent; Supplier Partnership Agent; Entitlement Service; Membership Repository; Entitlement Catalogue; Agent Trace

Preconditions:

- UC-05 has confirmed a request that requires a trip, plan, Premium Feature, or member-offer entitlement decision.
- The Operations Agent can access the customer, subscription, plan, and entitlement records.
- The membership, trip-count, trip-duration, advertising, feature-access, and member-offer eligibility rules are configured.
- The Agent Trace is available.

Success Condition: UC-07 records and returns Allowed or Denied with the verified membership status, applied rule, reason, and permitted next action. An allowed request continues through UC-05.

Failure Condition: If required membership evidence or the rule service is unavailable, UC-07 returns Verification-Failed, records the service failure, and blocks the protected action. Cases that meet the UC-05 escalation conditions are sent to Human Assistance.

Trigger: A Traveller requests creation or extension of a trip, activation of a Plan Version, access to a Premium Feature, or application of an approved member offer through the controlled Operations interface.

Main Success Scenario

1. UC-05 confirms a request that requires creation or extension of a trip, activation of a Plan Version, access to a Premium Feature, or application of an approved member offer.
2. The Operations Agent invokes UC-07 with the verified customer identifier, requested operation, trip dates when applicable, and related Case.
3. UC-07 retrieves the current membership status, active subscription evidence, Planned and Active trip count, and applicable feature or member-offer entitlements.
4. The deterministic Entitlement Service evaluates the request against the current Free Member or Paid Member rules.
5. UC-07 records the initial decision, membership snapshot, applied rule, and reason in the Agent Trace.
6. If travel planning is allowed, UC-05 routes the request to the Travel Planning Agent through UC-01.
7. UC-01 creates the Plan Version and returns its status, start date, end date, calculated calendar-day duration, and other required plan facts.
8. The Operations Agent invokes UC-07 for the post-planning review.

9. UC-07 retrieves the latest verified membership status and checks the Plan Version against the applicable trip-count, trip-duration, feature, and member-offer rules.
10. UC-07 records the final decision, applied rule, reason, and Plan Version.
11. UC-05 communicates the decision, reason, and permitted next action to the Traveller.

Alternative and Error Flows

A1. Free Member trip-count limit is reached (at Steps 4 or 9)

If the Free Member already has two simultaneous trips in Planned or Active status, UC-07 returns Denied, identifies the rule that failed, and preserves all existing trips. UC-05 explains that the Traveller may complete an existing trip or upgrade before trying again.

A2. Free Member trip duration exceeds five calendar days (at Step 9)

UC-01 supplies the start date, end date, and duration calculated as end date - start date + 1 calendar day. If the duration exceeds five calendar days, UC-07 returns Denied. UC-05 may offer the Traveller the valid choices of shortening the requested trip or upgrading; UC-07 does not change the itinerary.

A3. Membership changes during planning (at Steps 8-9)

UC-07 uses the latest verified membership status during the post-planning review. If it differs from the initial snapshot, UC-07 records both snapshots and evaluates the current request using the latest verified status.

A4. Membership evidence cannot be verified (at Steps 3-4 or 9)

UC-07 returns VerificationFailed, records the unavailable evidence or rule service, and blocks the protected action. The Case remains open under UC-05, which sends it to Human Assistance when its escalation conditions are met.

A5. Premium Feature access is requested (at Steps 3-4)

Interactive Route Map is available to active Paid Members. Other Premium Features are checked against the Entitlement Catalogue. If the membership or feature entitlement does not allow access, UC-07 returns Denied with the applied rule and next permitted action. UC-07 records and returns the decision through UC-05 without performing the trip-planning steps.

A6. Advertising or an approved member offer is evaluated (at Steps 3-4)

Commercial advertising shown to Free Members must be clearly labelled and must not affect hard-constraint filtering or recommendation ranking. Active Paid Members do not receive commercial advertising. For a member offer defined and approved through UC-08, UC-07 records and returns only the eligibility decision and applied membership level; it does not define the discount, calculate the quotation, or perform the trip-planning steps.

A7. The Travellerdisputes the decision (after Step 11)

A deterministic denial is returned with its rule and explanation. Conflicting evidence, unresolved uncertainty, or the UC-05 mandatory escalation condition sends the Case to Human Assistance. UC-07 does not silently override the Entitlement Service.

Business Rules

1. A Free Member may hold no more than two simultaneous trips whose status is Planned or Active.
2. A Free Member trip may cover no more than five calendar days.
3. Trip duration is calculated as end date - start date + 1 calendar day.
4. Active Paid Members are not subject to the Free Member trip-count or trip-duration limits.
5. Active Paid Members receive an ad-free interface, Interactive Route Map access, and other approved Premium Features.
6. Advertising shown to Free Members must be clearly labelled and cannot alter hard-constraint filtering or recommendation ranking.
7. The Entitlement Service returns Allowed, Denied with a reason, or VerificationFailed.
8. UC-01 calculates itinerary dates, duration, and feasibility. UC-07 evaluates the returned plan facts without modifying the itinerary.
9. 9. UC-08 defines an approved MemberOfferRule. UC-07 evaluates membership eligibility. UC-06 calculates a separate customer-specific PriceBreakdown using the approved MemberOfferRule and the UC-07 eligibility decision, then rechecks the final selling price and minimum margin without overwriting the reusable SupplierOffer base price.
10. The Operations Agent may use the LLM to identify the requested capability and explain the decision. Deterministic rules make the entitlement decision.
11. UC-07 does not process payments, refunds, subscription transactions, or itinerary changes.
12. UC-07 records the Case, Plan Version when applicable, membership snapshot, current trip count, decision, applied rule, reason, and timestamp in the Agent Trace.

Tools and Data

- Authentication and Access Control mechanism
- Customer and Membership Repository
- Plan Repository
- Entitlement Catalogue
- Entitlement and Policy Rules
- Plan facts supplied by UC-01
- Member offer rules supplied by UC-08

- Membership eligibility result supplied to UC-06
- Human Assistance Queue
- Agent Trace

Acceptance Criteria

AC-UC07-01 - Initial entitlement check

Given a confirmed request, when the Operations Agent invokes UC-07, then the system records and returns the current membership level, decision, applied rule, and reason.

AC-UC07-02 - Free Member trip-count limit

Given a Free Member with two simultaneous trips in Planned or Active status, when another trip requests activation, then UC-07 returns Denied, identifies the failed trip-count rule, and preserves the existing trips.

AC-UC07-03 - Free Member trip-duration limit

Given a Free Member Plan Version covering more than five calendar days, when UC-07 performs the post-planning review, then it returns Denied with the duration rule and does not modify the itinerary.

AC-UC07-04 - Post-planning review

Given a Plan Version returned by UC-01, when UC-07 performs the post-planning review, then it uses the latest verified membership status, current trip count, and calculated calendar-day duration and records whether the Plan may proceed.

AC-UC07-05 - Premium Feature access

Given a request for Interactive Route Map or another registered Premium Feature, when UC-07 checks the Entitlement Catalogue, then it returns the access decision, membership level, and entitlement used.

AC-UC07-06 - Approved member offer

Given an approved UC-08 member-offer rule, when eligibility is checked, then UC-07 returns the membership eligibility result and applied level without defining or changing the discount.

AC-UC07-07 - Verification failure

Given that membership evidence or the rule service is unavailable, when UC-07 checks a protected request, then it returns VerificationFailed, records the service failure, blocks the protected action, and follows the UC-05 escalation rules.

7.3 For the Travel Planning Agent

7.3.1 UC-01: Create and Compare a Travel Plan

Owner / Author: Mo Shen
Related requirements: TP-01, TP-AH-01
Related diagrams: AD-02, SD-02, SM-02, G-03

Name: Create and Compare a Travel Plan

Goal: Produce complete and feasible itinerary options from a confirmed request, recommend one option, and compare it with eligible alternatives.

Level: User-goal level

Primary Actor: Traveller

Supporting External Actor: Human Founder/Operator

Internal Participants: Operations Agent; Travel Planning Agent; Supplier Partnership Agent; Reviewer/Guardrail; deterministic validation services; Agent Trace

Preconditions:

1. UC-05 has verified and confirmed the planning request.
2. The required travel fields and UC-07 entitlement result are available.
3. Approved supplier data and deterministic validation services are available.

Success Condition: Operations receives a validated recommended itinerary and eligible alternatives with cost breakdowns, evidence, trade-offs, and comparison rationale recorded in the Agent Trace.

Failure Condition: No plan is represented as feasible when hard constraints cannot be met, required evidence cannot be verified, or a required approval remains unresolved; the Case remains open for clarification or human handling.

Trigger: The Traveller submits and confirms a new travel-planning request through the controlled Operations interface.

Main Success Scenario

1. Operations sends the confirmed request, Traveller profile, entitlement result, Case identifier, and unresolved questions to Travel Planning.
2. Travel Planning normalises the request into a versioned constraint set.
3. Travel Planning retrieves approved supplier data and records each source, offer identifier, retrieval time, validity, availability, price, cancellation terms, and reliability information.
4. Travel Planning assembles complete door-to-door itinerary candidates.
5. Deterministic services validate dates, budget, schedule feasibility, transfer time, opening hours, accessibility, baggage, availability, cancellation status, and policy restrictions.
6. The services calculate itemised and total prices and reject infeasible, unavailable, conflicting, stale, or unsupported candidates.
7. Travel Planning ranks only feasible candidates in the confirmed Comfort, Value, or Budget mode.
8. Travel Planning prepares a recommended itinerary and eligible alternatives with cost breakdowns, constraint coverage, sources, timestamps, trade-offs, and exclusion reasons.
9. Reviewer/Guardrail checks the evidence, calculations, policy compliance, and approval boundary.
10. Travel Planning records and returns the reviewed result to Operations for controlled communication.

Alternative and Error Flows

A1. Missing or ambiguous preference (at Steps 1-2)

Travel Planning returns focused clarification questions through Operations. Processing resumes after the Traveller confirms the missing information.

A2. No feasible candidate (at Steps 5-7)

Travel Planning returns the unmet hard constraints, closest valid alternatives, and explicit relaxation choices. A hard constraint is not relaxed without Traveller confirmation through Operations.

A3. Stale, missing, or conflicting supplier data (at Steps 3-6)

The affected offer is excluded, the evidence issue is recorded, and refreshed information is requested through the appropriate participant. The offer is not presented as confirmed while the issue remains unresolved.

A4. Restricted commitment or unresolved low confidence (at Steps 8-10)

Travel Planning returns the evidence, assumptions, risks, and options to Operations for Human Founder/Operator review. The result remains pending until the required decision is recorded.

Business Rules and Approval Conditions

1. Every hard constraint must pass before ranking.
2. Deterministic validation and calculations override inconsistent LLM suggestions.
3. Sources and retrieval timestamps accompany planning claims.
4. Comfort, Value, or Budget mode affects ranking only after feasibility checks pass.
5. Operations controls communication with the Traveller.
6. Financial, contractual, safety, legal, or policy-sensitive commitments require Human Founder/Operator approval.

Tools and Data

- Confirmed request and Traveller profile from UC-05
- Entitlement result from UC-07
- Approved SupplierOffer records
- Constraint and schedule validators
- Price calculator
- Reviewer/Guardrail
- TravelPlan Repository
- Agent Trace

Acceptance Criteria

AC-UC01-01 — Complete itinerary: Given a confirmed planning request, when a plan is returned as feasible, then it contains the required travel legs, accommodation, and selected activities.

AC-UC01-02 — Hard constraints: Given an itinerary candidate, when validation is performed, then every applicable hard constraint has a recorded pass before ranking.

AC-UC01-03 — Price reconciliation: Given a returned itinerary, when its price is reviewed, then the total reconciles to its recorded components.

AC-UC01-04 — Comparison: Given multiple feasible candidates, when the recommendation is returned, then the alternatives use common comparison criteria and state their trade-offs.

AC-UC01-05 — No feasible plan: Given that no candidate satisfies the confirmed hard constraints, when planning ends, then no plan is labelled feasible and the unmet constraints and permitted next actions are returned.

7.3.2 UC-02: Revise Preferences and Compare Versions

Owner / Author: Mo Shen
Related requirements: TP-02, TP-AH-01
Related diagrams: AD-02, SD-02, SM-02, G-03

Name: Revise Preferences and Compare Versions

Goal: Apply a Traveller-confirmed preference or constraint change to a copy of an existing TravelPlan, validate the revised options, and present a transparent comparison with the baseline.

Level: User-goal level

Primary Actor: Traveller

Supporting External Actor: Human Founder/Operator

Internal Participants: Operations Agent; Travel Planning Agent; Supplier Partnership Agent; Reviewer/Guardrail; deterministic validation services; Agent Trace

Preconditions:

1. The authenticated Traveller can access the baseline TravelPlan.
2. Operations has captured and confirmed the requested amendment.
3. The baseline constraints, evidence, and version history are available.
4. Approved supplier data and deterministic validation services are available.

Success Condition: A new immutable TravelPlan version and field-level comparison with the baseline are returned to Operations and recorded.

Failure Condition: The baseline remains unchanged when the amendment is unconfirmed, invalid, infeasible, unsupported, or awaiting a required approval.

Trigger: The Traveller requests a change to the dates, budget, accommodation, accessibility, arrival time, transfer tolerance, activity choice, or cancellation preference of an existing TravelPlan.

Main Success Scenario

1. Operations sends the authenticated request, baseline version identifier, confirmed amendment, entitlement result, and Case context to Travel Planning.

2. Travel Planning loads the baseline plan, constraints, offer evidence, validation result, and rationale.
3. Travel Planning identifies the changed fields and confirms that the amendment is complete and consistent.
4. Travel Planning creates a draft successor version while preserving the baseline.
5. Travel Planning refreshes affected supplier offers and reconstructs the affected itinerary elements.
6. Deterministic services recalculate schedules, transfers, availability, cancellation conditions, component costs, total cost, and policy compliance.
7. Travel Planning ranks the feasible revised candidates.
8. Travel Planning compares the draft with the baseline, including changed requirements, resources, schedule, price delta, constraint coverage, evidence freshness, assumptions, and trade-offs.
9. Reviewer/Guardrail checks the comparison, calculations, evidence, and approval boundary.
10. Travel Planning records the new version, comparison, validation outcome, and rationale and returns the reviewed result to Operations.

Alternative and Error Flows

B1. Incomplete amendment (at Steps 1-3): Operations obtains focused clarification; the baseline remains current until the Traveller confirms the amendment.

B2. Conflict with an existing hard constraint (at Steps 3 or 6): Travel Planning identifies the conflict and Operations obtains an explicit Traveller choice. No hard constraint is silently replaced.

B3. No feasible revision (at Steps 6-7): The baseline remains current, and Travel Planning returns the failed constraints, closest valid alternatives, or explicit relaxation choices.

B4. Stale or inconsistent evidence (at Steps 5-6): The affected evidence is flagged and refreshed. The revision is not presented as confirmed while the issue remains unresolved.

B5. Approval-bound change (at Steps 8-10): Travel Planning creates an escalation package and Operations waits for the Human Founder/Operator decision.

Business Rules and Approval Conditions

1. Every confirmed amendment creates a new immutable version.
2. The baseline is never overwritten.
3. Baseline and revision use the same comparison fields and calculation rules.
4. Traveller confirmation is recorded before a revised plan is released.
5. A restricted commitment requires Human Founder/Operator approval.

Tools and Data

- Confirmed amendment and Case context from UC-05
- Baseline TravelPlan and version history
- Entitlement result from UC-07
- Approved SupplierOffer records
- Deterministic validation and price services
- Reviewer/Guardrail
- Agent Trace

Acceptance Criteria

AC-UC02-01 — Version preservation: Given a confirmed amendment, when a revision is created, then the baseline remains unchanged and both versions have distinct identifiers.

AC-UC02-02 — Transparent changes: Given a baseline and revised version, when comparison is returned, then every changed requirement, itinerary resource, schedule field, and price delta is visible.

AC-UC02-03 — Revalidation: Given a revised candidate, when it is returned as feasible, then its price and applicable hard constraints have been recalculated and recorded.

AC-UC02-04 — Infeasible amendment: Given an amendment that cannot satisfy the confirmed constraints, when revision processing ends, then the baseline remains current and the failed constraints are explained without silent substitution.

7.4 For the Customer Exception Agent

7.4.1 UC-03: Recheck an Active Trip and Recover from Disruption

Owner / Author: Runda Weng
Related requirements: EXC-01, EXC-AH-01
Related diagrams: AD-03, SD-03, SM-03, G-03

Name: Recheck an Active Trip and Recover from Disruption

Goal: Reassess an active trip affected by a disruption and produce a validated recovery proposal.

Level: User-goal level

Primary Actor: Traveller

Supporting External Actor: Human Founder/Operator

Internal Participants: Customer Exception Agent; Operations Agent; Travel Planning Agent; Supplier Partnership Agent; Reviewer/Guardrail; deterministic validation tools; Agent Trace

Preconditions:

1. UC-05 has completed the required identity and access checks and confirmed the disruption-related request.
2. The affected active itinerary is available for investigation.

Success Condition: A recovery proposal has passed the relevant validation and review checks and has been returned to Operations with supporting evidence and any approval requirements.

Failure Condition: When a validated recovery proposal cannot be produced, the available findings, missing information, and escalation reasons are recorded, and the Case remains open for clarification or human handling.

Trigger: The Traveller reports a disruption affecting an active trip through the controlled Operations interface.

This use case produces a recovery proposal; it does not execute booking changes, payments, refunds, or compensation. Operations manages communication and coordinates any required approval through UC-05.

Main Success Scenario

1. The Operations Agent passes the confirmed disruption-related request and relevant Traveller context to the Customer Exception Agent.
2. The Customer Exception Agent reviews the active itinerary and gathers evidence about the reported disruption.
3. The agent identifies the affected travel arrangements and assesses their implications for the remaining itinerary.
4. The agent identifies the relevant recovery constraints, including the Traveller's preferences, budget, and timing.
5. The Supplier Partnership Agent provides information on suitable alternative services, availability, prices, and supplier terms.
6. The Travel Planning Agent uses the available alternatives and relevant constraints to prepare revised itinerary options.
7. The Customer Exception Agent consolidates the findings into a recovery proposal explaining the proposed changes and their implications.
8. Deterministic tools validate the relevant constraints and calculations. The shared Reviewer/Guardrail checks the supporting evidence, policy compliance, and approval requirements.
9. The Customer Exception Agent returns the validated proposal, supporting evidence, and identified approval requirements to the Operations Agent.
10. The system records the investigation, tool results, proposal, review outcome, and hand-off. UC-05 continues with response review, any required human approval, and Traveller communication.

Alternative and Error Flows

A1. Additional information is required (Steps 2-4)

If itinerary details or supporting evidence are insufficient, the Customer Exception Agent identifies the missing information and requests clarification through the Operations Agent. The case remains pending until the information is available or human handling is required.

A2. Evidence is conflicting or confidence remains insufficient (Steps 2-3)

The agent records the conflicting information and seeks clarification. If the uncertainty cannot be resolved sufficiently to support recovery planning, the available findings and unresolved issues are referred for human handling.

A3. No suitable alternative is available (Steps 5-6)

If no available option satisfies the relevant constraints, the agent records the reasons and returns an escalation recommendation to the Operations Agent for referral to the Human Founder/Operator.

A4. A specialist Agent or required tool is unavailable (Steps 2-8)

The agent records the unavailable dependency and incomplete work. The Operations Agent is informed so that the case remains open and receives human assistance under UC-05.

A5. Validation or review fails (Step 8)

The proposal is returned to the responsible specialist with the failure reasons. Revised proposals must pass the relevant checks. If the issues cannot be resolved, the case is referred for human handling.

A6. Human approval is required (Steps 8-10)

The proposal is marked with the relevant approval requirements. The Operations Agent obtains the Human Founder/Operator's decision through UC-05 before communicating a restricted commitment. A proposal requiring revision is returned with the review reasons, and a rejected commitment is not presented as approved.

A7. A legal or safety concern is identified (at any step)

The concern and available evidence are recorded, and the matter is referred to the Human Founder/Operator under the shared oversight rules.

Postconditions and Minimum Guarantee

The use case succeeds when a recovery proposal has passed the relevant validation and review checks and has been returned to the Operations Agent with supporting evidence and any approval requirements. The proposal and its review results are recorded.

If a validated recovery proposal cannot be produced, the available findings, missing information, and escalation reasons are recorded. The case remains awaiting further information or human handling and is not represented as successfully recovered.

In either outcome, producing a recovery proposal does not itself change a booking or execute a financial transaction.

Business Rules and Approval Conditions

The Operations Agent is the controlled gateway for communication with the Traveller. Recovery proposals must consider the relevant itinerary constraints and use validated supplier information and calculations. LLM-supported analysis does not replace deterministic validation or authorised human decisions.

Required validation and review checks must pass before a proposal is represented as validated. Payments, refunds, compensation, binding supplier commitments, and other restricted actions require human approval before commitment or execu-

tion. Legal or safety concerns and unresolved low-confidence matters require human handling.

Requests, evidence, tool results, itinerary versions, proposals, reviews, approvals, and hand-offs must remain traceable. Supplier, booking, payment, and disruption interactions use simulated data within the teaching prototype.

Tools and Data

The use case requires the confirmed Traveller request and context from UC-05, the active itinerary version, relevant Traveller constraints, disruption information, and supporting evidence. Recovery planning also uses supplier offers, availability, prices, terms, and revised itinerary options from the Travel Planning Agent.

Supporting capabilities include deterministic constraint and calculation validators, the shared Reviewer/Guardrail, applicable policies, Agent Trace, and human approval records.

Acceptance Criteria

AC-UC03-01 — Investigation: Given a confirmed disruption request and an accessible active itinerary, when the investigation is performed, then the affected arrangements and supporting evidence are recorded.

AC-UC03-02 — Recovery proposal: Given suitable alternatives, when a recovery proposal is prepared, then it identifies the proposed changes and their relevant budget, timing, and preference implications.

AC-UC03-03 — Validation: Given a recovery proposal, when it is returned as validated, then the relevant validation and review results are recorded as passed.

AC-UC03-04 — Incomplete processing: Given insufficient information, unavailable dependencies, or no suitable alternative, when processing cannot continue, then the reason and available findings are recorded and passed for clarification or human handling.

AC-UC03-05 — Approval: Given a restricted commitment, when the proposal is handed to Operations, then the approval requirement is explicit, and the commitment is not communicated as approved without a recorded human decision.

AC-UC03-06 — Traceability: Given completed specialist processing, when the result is handed to Operations, then the request, evidence, tool results, proposal, and review outcome are linked in the trace record.

7.4.2 UC-04: Investigate a Complaint and Recommend Resolution

Owner / Author: Runda Weng
Related requirements: EXC-02, EXC-AH-02
Related diagrams: AD-03, SD-03, SM-03, G-03

Name: Investigate a Complaint and Recommend Resolution

Goal: Investigate a confirmed Traveller complaint and produce an evidence-supported resolution recommendation or a documented referral for human handling.

Level: User-goal level
Primary Actor: Traveller
Supporting External Actor: Human Founder/Operator
Internal Participants: Customer Exception Agent; Operations Agent; Travel Planning Agent; Supplier Partnership Agent; Reviewer/Guardrail; deterministic rule and calculation tools; Agent Trace

Preconditions:

1. UC-05 has completed the required identity and access checks and confirmed the complaint.
2. The provided Traveller context is available to the Customer Exception Agent.

Success Condition: A reviewed resolution recommendation or documented human referral is returned to Operations with supporting findings and any approval requirements.

Failure Condition: If neither a reviewed recommendation nor a complete referral can be produced, the incomplete work and reasons are recorded and the Case remains open.

Trigger: The Traveller submits and confirms a complaint through the controlled Operations interface.

Operations manages communication with the Traveller and coordinates approvals through UC-05. A recommendation does not itself authorise a refund, compensation, or other binding commitment. A human referral completes the specialist hand-off but does not mean the underlying complaint has been resolved.

Main Success Scenario

1. The Operations Agent passes the confirmed complaint and relevant Traveller context to the Customer Exception Agent.
2. The Customer Exception Agent reviews the complaint and identifies the issue requiring investigation.
3. The agent gathers relevant evidence, including available itinerary information, service records, Traveller-provided material, and applicable policies or supplier terms.
4. The agent obtains relevant specialist findings. The Travel Planning Agent assists with itinerary-related issues, while the Supplier Partnership Agent assists with supplier services, prices, and terms.
5. The Customer Exception Agent assesses the evidence against the reported issue and relevant rules.
6. The agent prepares a resolution proposal explaining the findings, supporting evidence, recommended response, and any approval requirements.
7. The shared Reviewer/Guardrail checks the proposal against the relevant evidence, policies, calculations, and approval rules.
8. The Customer Exception Agent returns the reviewed proposal and supporting findings to the Operations Agent.
9. The system records the investigation, evidence, specialist findings, review outcome, and hand-off. UC-05 continues

with response review, required human approval, and communication with the Traveller.

Alternative and Error Flows

A1. Complaint details or evidence are insufficient (Steps 2-5)

The agent identifies the missing information and requests clarification through the Operations Agent. The case remains pending while additional information is required. If the matter cannot be clarified, the available evidence and unresolved questions are referred for human handling.

A2. Evidence conflicts or does not support a confident recommendation (Steps 3-5)

The agent records the conflicting or inconclusive findings. It seeks relevant clarification and, if uncertainty remains unresolved, prepares a documented human referral instead of presenting an unsupported resolution.

A3. A required record, tool, or specialist is unavailable (Steps 3-4)

The agent records the unavailable dependency and its effect on the investigation. The Operations Agent is informed, and the case remains open for further information or human assistance under UC-05.

A4. Review requires revision (Step 7)

The proposal is returned with the review reasons. The agent revises the proposal and submits it for the relevant checks again. Unresolved issues are referred for human handling.

A5. The proposed resolution requires human approval (Steps 6-9)

Refunds, compensation, and other restricted commitments are explicitly marked as requiring approval. The Operations Agent obtains the Human Founder/Operator's decision through UC-05 before communicating the commitment. Rejected commitments are not represented as approved.

A6. The complaint requires human judgement (at any step)

Where the complaint involves legal or safety concerns, unresolved low confidence, or a decision outside the agent's authority, the agent prepares a referral containing the complaint, evidence, findings, and reason for escalation. The Operations Agent coordinates the referral to the Human Founder/Operator.

Postconditions and Minimum Guarantee

The use case succeeds when a reviewed resolution proposal or a documented human escalation is returned to the Operations Agent with supporting findings and any approval requirements.

For the resolution-proposal outcome, the recommendation, supporting evidence, and review results are recorded. For the escalation outcome, the complaint context, findings, unresolved issues, and reason for human handling are recorded and handed over. The underlying complaint remains subject to the human decision.

If neither a reviewed proposal nor a completed referral is available, the incomplete work and reasons are recorded, and the case remains open. No refund, compensation, or binding commitment is executed by this use case.

Business Rules and Approval Conditions

The Operations Agent manages communication with the Traveller. Complaint recommendations must be supported by relevant evidence and applicable policies or supplier terms. Unresolved evidence gaps and uncertainty must be recorded and included in any human referral.

LLM-supported interpretation does not replace deterministic financial or contractual rules or authorised human judgement. Resolution proposals must undergo the relevant review before being returned for Traveller communication.

Refunds, compensation, financial commitments, binding commercial terms, and other restricted commitments require human approval. Complaints involving legal or safety concerns or unresolved low confidence are referred to the Human Founder/Operator. The Operations Agent coordinates approval and external communication through UC-05.

Evidence, specialist findings, recommendations, review outcomes, approvals, and hand-offs must remain traceable. Financial and supplier interactions remain simulated within the teaching prototype.

Tools and Data

The use case requires the confirmed complaint and Traveller context from UC-05, relevant itinerary and service records, Traveller-provided evidence, and applicable policies and supplier terms. It also uses relevant findings from the Travel Planning and Supplier Partnership Agents.

Supporting capabilities include deterministic rule and calculation tools, the shared Reviewer/Guardrail, Agent Trace, the human assistance queue, and approval records.

Acceptance Criteria

AC-UC04-01 — Complaint intake: Given a confirmed complaint, when investigation begins, then the reported issue and available Traveller context are recorded.

AC-UC04-02 — Evidence-supported recommendation: Given available evidence and specialist findings, when a resolution is proposed, then the recommendation identifies its supporting evidence and relevant policies or terms.

AC-UC04-03 — Review: Given a resolution proposal, when it is returned as reviewed, then the relevant Reviewer/Guardrail results are recorded as passed.

AC-UC04-04 — Uncertainty: Given missing or conflicting evidence, when a confident recommendation cannot be established, then the unresolved issues and available findings are recorded and passed for clarification or human handling.

AC-UC04-05 — Approval: Given a proposed refund, compensation, or other restricted commitment, when the proposal is handed to Operations, then the approval requirement is explicit, and the commitment is not communicated as approved without a recorded human decision.

AC-UC04-06 — Human escalation: Given a case requiring human handling, when the referral is completed, then the complaint, evidence, findings, and escalation reason are available to the Human Founder/Operator through the coordinated hand-off.

AC-UC04-07 — Traceability: Given completed specialist processing, when the outcome is handed to Operations, then the complaint, evidence, recommendation or referral, and review outcome are linked in the trace record.

7.5 For the Supplier Partnership Agent

7.5.1 UC-06: Source Supplier Inventory and Validate Margin

Owner / Author: Jason Yang
Related requirements: SUP-01, SUP-02, SUP-AH-01
Related diagrams: AD-04, SD-04, SM-04, G-03

Name: Source Supplier Inventory and Validate Margin

Goal:

Assess a supplier offer and return an approved SupplierOffer and PriceBreakdown for internal planning.

Level: User-goal level

Primary Actor: Travel Service Provider

Supporting External Actor: Human Founder/Operator

Internal Participants: Operations Agent; Supplier Partnership Agent; LLM extraction function; deterministic validation and pricing functions; Reviewer/Guardrail; Agent Trace

Trigger:

A Travel Service Provider submits and confirms a supplier offer or correction through the controlled Operations interface.

Preconditions:

Operations has performed the required intake/access checks and confirmed the supplied information under UC-05.

The prototype contains simulated supplier data and configured validation, markup, minimum-margin and approval rules.

Specialist validation may still fail after intake. Service availability is not assumed to guarantee success.

Success Condition:

The offer is Approved with a saved PriceBreakdown, Reviewer pass and any required human approval.

The outcome and key decisions are saved and returned to Operations. Only approved and unexpired offers are selectable.

Operations retains responsibility for external delivery and SupplierCase closure.

Failure Condition:

Failed information, validation, margin, or review checks leave the offer RevisionRequired. A human rejection produces Rejected. Pending approval leaves AwaitingApproval. A tool or storage error retains the current state, prevents the offer from becoming available, and allows manual retry. Failed assessment does not close the Operations Case.

Main Success Scenario

1. Operations forwards the confirmed case and supplier source.
2. Supplier Partnership creates or updates a Draft offer; the LLM extracts fields, source evidence and uncertainties.
3. Validation/Pricing functions validate fields, supplier eligibility, availability, expiry, price basis and terms.
4. The functions calculate selling price and margin, check the minimum margin and return a PriceBreakdown.
5. The offer enters UnderReview. Reviewer checks evidence, calculations and rules and returns a pass and the applicable approval requirement.
6. When required by the configured approval rules, the offer enters AwaitingApproval and the Human Founder/Operator approves it. If human approval is not required, processing continues after the Reviewer pass.
7. Supplier Partnership saves the Approved outcome, PriceBreakdown and key trace records. A save failure does not make the offer available.
8. Supplier Partnership returns the result to Operations; UC-05 handles the external response.

Alternative and Error Flows

A1. Information, validation, or margin failure (Steps 2-4)

Save RevisionRequired with the specific issue and return it to Operations. Operations coordinates a permitted correction and obtains confirmation for supplier facts. On a confirmed correction, clear the previous review and approval, set Draft, and repeat Steps 2-7. The Agent cannot invent supplier facts or override a failed rule. If no correction is supplied, the offer remains unavailable; it need not automatically time out or be reassigned.

A2. Reviewer failure (Step 5)

Return RevisionRequired with the failed check and follow A1. No failed review can proceed to approval.

A3. Human decision (Step 6)

Approve continues to save. Request revision follows A1. Reject saves Rejected and the reason and returns the outcome to Operations. No decision leaves AwaitingApproval. An approval cannot override failed validation or margin checks.

A4. Expiry or material correction after approval

Check the current time against validUntil whenever an offer is selected. An expired offer is unavailable and is marked Expired during that read. A material correction moves the offer to RevisionRequired, clears approval and repeats assessment after confirmation. The prototype does not need a background expiry service or notifications to other Agents. Confirmed bookings are outside this use case.

A5. LLM, function, or save error

Show the error, keep the existing business state, and do not return an approved result. Operations or the Human Founder/Operator can retry manually.

A6. External delivery fails

Operations keeps the case open and handles the failure under UC-05. Supplier Partnership does not implement a second delivery system.

A7. Member-offer quotation

When an approved member offer is requested, obtain the UC-07 eligibility result and check the rule's validity and applicability. If eligibility cannot be verified, pause the quotation. If the member is ineligible, do not apply the discount and explain the result through Operations. If eligible, calculate a separate customer-specific PriceBreakdown using the recorded fixed discount and repeat the price and minimum-margin checks without modifying the reusable SupplierOffer base price. A failed margin check blocks the discounted quotation and returns it for revision; the Agent cannot reduce the supplier cost or override the rule.

Business Rules and Approval Conditions

Operations owns all external messages and supplier confirmation.

LLM extraction/explanation does not replace deterministic validation or arithmetic.

supplierCost, basePrice, discountAmount, sellingPrice, and marginAmount use the same quoted quantity and AUD basis. Supplier cost includes known mandatory charges; unknown values must be corrected.

basePrice = round(supplierCost x (1 + markupRate), 2); sellingPrice = round(basePrice - discountAmount, 2); marginAmount = sellingPrice - supplierCost. A base offer uses discountAmount = 0. The selling price must be positive, and the configured minimum margin must pass after any approved discount.

Reviewer pass is mandatory. Human approval is required for binding prices, agreements, restricted terms, financial commitments and any other configured restricted decision.

A material revision clears the previous approval and repeats assessment. Saving an approval does not execute a transaction.

Only Approved and unexpired offers are selectable. Tool errors and incomplete records are not approval.

Tools and Data

LLM extraction function for structured offer fields, supporting evidence, and uncertainty.

Deterministic validation and pricing functions for a passed PriceBreakdown or a failure reason.

Reviewer/Guardrail result containing pass or failure, reasons, and whether human approval is required.

Human decision form: Approve / Request revision / Reject and reason.

Offer Repository and Agent Trace containing the offer ID, Case ID, source, fields, status, breakdown, review, approval, and timestamped events.

Acceptance Criteria

AC-UC06-01: A missing field, invalid supplier, unavailable item or expired offer does not reach approval; the reason is returned through Operations.

AC-UC06-02: With supplierCost = AUD 180.00, markupRate = 10%, and minimum margin = AUD 15.00, the calculation returns sellingPrice = AUD 198.00 and marginAmount = AUD 18.00. With a 5% markup, sellingPrice = AUD 189.00 and marginAmount = AUD 9.00, so the margin check fails. These are test fixtures, not agreed business defaults.

AC-UC06-03: Reviewer failure produces RevisionRequired and no approved output.

AC-UC06-04: A restricted offer without a human decision remains AwaitingApproval. Approve, Request revision and Reject have the three specified outcomes.

AC-UC06-05: A confirmed correction clears old decisions and repeats extraction/checks before approval.

AC-UC06-06: A successfully saved approved result includes the breakdown and trace; a save/tool error cannot produce an approved output.

AC-UC06-07: Reading an expired Approved offer excludes it from selection. No scheduler is needed to pass this check.

AC-UC06-08: Every supplier-facing message passes through Operations, which retains case delivery and closure responsibilities.

AC-UC06-09: Given supplierCost = AUD 180.00, basePrice = AUD 198.00, an approved discount of AUD 5.00 and an eligible member, the system calculates sellingPrice = AUD 193.00 and marginAmount = AUD 13.00. A configured minimum margin of AUD 15.00 therefore fails, and the discounted quotation is not approved. These values are test fixtures.

Assumptions and boundaries

The prototype uses simulated supplier data, AUD prices, a consistent quoted quantity and two-decimal rounding. Required offer fields, markupRate, minimum margin, and approval conditions are configuration values rather than fixed business defaults in this requirement. Acceptance-criteria amounts are test fixtures only. UC-08 owns campaign and MemberOfferRule design, UC-07 evaluates membership eligibility, and UC-06 calculates a separate customer-specific PriceBreakdown using the approved MemberOfferRule and UC-07 eligibility decision. UC-06 rechecks the final selling price and minimum margin without overwriting the reusable SupplierOffer base price. A failed discounted quotation does not invalidate an otherwise valid approved SupplierOffer. Real purchasing, booking, and payment execution are outside this prototype. Authentication, intake, and external delivery remain the responsibility of UC-05.

7.5.2 UC-08: Plan, Evaluate, and Approve a Marketing Campaign

Owner / Author: Jason Yang
Related requirements: SUP-03, SUP-AH-02
Related diagrams:

G-03 (overall use-case coverage). AD-04, SD-04, SM-04 and G-05A/B/C describe the UC-06 supplier-offer assessment dependency; they do not directly model the UC-08 campaign approval workflow.

Name: Plan, Evaluate, and Approve a Marketing Campaign

Goal:

Prepare a supplier-data-based marketing campaign, evaluate its evidence and budget, and obtain a recorded Human Founder/Operator decision before it can be used for an authorised campaign action.

Level: User-goal level

Primary Actor: Human Founder/Operator, who defines the campaign brief and approves, requests revision, or rejects the proposal.

Supporting External Actor: Travel Service Provider, only when supplier clarification is needed through Operations. A real Advertising Channel is relevant to a later explicitly authorised publication action; real channel integration is outside the base prototype.

Internal Participants: Supplier Partnership Agent; Operations Agent; LLM drafting function; approved supplier-offer data; deterministic campaign evaluation functions; Reviewer/Guardrail; Campaign Repository; Agent Trace

Trigger:

The Human Founder/Operator submits a campaign brief requesting a proposal based on relevant supplier offers, or submits a revision to an existing brief.

Preconditions:

The operator has the required access to request and decide campaign proposals.

The system has configured permitted channels, content restrictions and a simulated cost basis for the selected channel.

Supplier data is accessible to the Agent. The existence of a suitable offer is not guaranteed; no suitable data follows A2.

The brief can be incomplete initially; missing information follows A1 rather than being guessed.

Success Condition:

An Approved campaign proposal is saved with the brief, selected supplier references, draft content, channel/dates, estimated cost breakdown, passed evaluation/review and human decision.

The proposal is available to the operator for preview or an authorised simulated campaign action.

No advertisement has been externally published and no real funds have been spent merely because the proposal was approved.

Failure Condition:

An incomplete or failed proposal stays Draft or RevisionRequired with reasons. A proposal awaiting a human decision stays AwaitingApproval. Human rejection produces Rejected. A tool or save failure prevents a successful approved result and supports manual retry. Unapproved or materially changed proposals cannot be used as approved campaign material.

Main Success Scenario

1. The operator provides the objective, audience, permitted channel, campaign dates, budget ceiling and content restrictions.
2. Supplier Partnership checks the brief for completeness and saves a Draft campaign with a campaign ID.
3. The Agent selects relevant Approved, unexpired supplier offers and records their prices, availability, validity, terms and source references. The campaign dates must fit the offers' applicable promotion validity; otherwise the proposal is revised.
4. The LLM drafts campaign content and a proposal explaining the selected offers, audience and channel in relation to the objective. Sources accompany factual claims.

If the campaign includes a member offer, the Agent records its offer ID, applicable supplier offers, eligible membership tiers, fixed discount amount and validity period. These terms form part of the proposal submitted for evaluation, review and human approval.

5. A deterministic function calculates the estimated campaign cost from configured simulated cost items, and checks the budget ceiling, permitted channel, campaign dates and consistency of advertised prices/terms with supplier data.
6. Reviewer/Guardrail checks evidence, claims, calculation results and configured content/privacy restrictions. On pass, the proposal enters AwaitingApproval and is presented to the operator with its cost breakdown and assumptions.
7. The operator approves the reviewed proposal. The Agent saves the decision, proposal and trace, and sets the campaign to Approved only after the save succeeds.
8. The system returns the approved proposal to the operator for preview. External publication and real spending are not executed by this base scenario.

Alternative and Error Flows

A1. Incomplete or ambiguous brief (Steps 1-2)

Ask the operator for the missing objective, dates, permitted channel, budget or restriction. Keep Draft until the brief is complete. The LLM must not invent a budget or permission to use a channel.

A2. No suitable supplier data (Step 3)

Explain why offers cannot be used, such as expiry, unavailable inventory, inconsistent price basis or unsuitable dates. Keep the campaign Draft or RevisionRequired and ask the operator to change the brief or provide suitable data. Supplier clarification goes through Operations. Do not fabricate a supplier offer or alter an approved price.

A3. Budget, dates, or evidence check fails (Steps 5-6)

Set RevisionRequired and record the failed checks. The Agent may draft a corrected proposal within the existing brief,

such as reducing planned campaign units. A higher budget ceiling or changed constraint requires operator confirmation. Repeat evaluation and review; do not silently relax the failed rule.

A4. Human decision (Step 7)

Approve proceeds to save. Request revision records the requested changes, clears prior approval and repeats the relevant drafting, evaluation and review steps. Reject saves Rejected and the reason. No decision leaves AwaitingApproval. Human approval cannot substitute for missing required evidence or a failed deterministic check; any permitted rule change must first be explicitly recorded and the proposal reassessed.

A5. Supplier data or campaign content changes

If selected offers become unsuitable, or campaign content, channel, dates, or budget change materially, mark RevisionRequired and clear prior approval. Refresh the evidence and repeat evaluation, review and human approval. The prototype can check this when the proposal is reopened or used; a background monitoring/notification service is not required.

A6. LLM, evaluation, or save error

Show the error and do not return a successfully approved campaign. Keep the current business state with an error and allow manual retry. Local campaign data and trace may be saved in a normal database transaction; no distributed recovery mechanism is required.

A7. Optional simulated campaign response after approval

If the prototype demonstrates a simulated campaign, a generated response records campaignId and sourceChannel and is sent to Operations. UC-05 confirms the request, routes it and controls external replies. A response does not transfer campaign design or budget ownership to Operations. This demonstration is optional to the base approval scenario.

A8. Request for real publication or spending

The base prototype does not execute the action. A future implementation must obtain explicit human authorisation for the publication/spending action and use an approved channel integration. Approval of a draft proposal must not be treated as blanket authorisation to spend or publish.

Business Rules

BR-08-01: Supplier Partnership owns supplier-data-based campaign planning, evaluation and approval coordination; Operations owns Campaign Response intake under UC-05.

BR-08-02: Use approved supplier evidence. The LLM may draft and explain but cannot invent prices, discounts, availability, permissions or guaranteed campaign outcomes.

BR-08-03: estimatedTotalCost = sum(plannedUnits x configuredUnitCost) + recordedFixedCosts. All items use the same AUD basis in the prototype. Count each charge once, and do not assume an unknown cost is zero.

BR-08-04: Estimated cost must not exceed the confirmed ceiling. Planned dates/channel and advertised prices/terms must pass configured checks.

BR-08-05: Reviewer pass and a recorded human approval are required for an Approved campaign. A material change invalidates that approval.

BR-08-06: Proposal approval, simulation, external publication and real spending are distinct actions. External publication and spending require explicit human authorisation.

BR-08-07: Record sources, assumptions, cost items, failed checks, revisions, reviews and decisions. Do not claim predicted clicks, conversions or revenue without an identified supporting basis; such prediction is not required here.

BR-08-08: An approved MemberOfferRule defines the applicable SupplierOffers, eligible membership tiers, fixed discount amount and validity period. UC-07 evaluates eligibility but does not design or change the discount. UC-06 calculates a separate customer-specific PriceBreakdown using the approved MemberOfferRule and UC-07 eligibility decision. It rechecks the final selling price and minimum margin without overwriting the reusable SupplierOffer base price. A failed discounted quotation does not invalidate an otherwise valid approved SupplierOffer. The prototype does not support discount stacking.

BR-08-09: Campaign audiences must respect UC-07 advertising rules. Paid Members must not receive commercial advertisements. Eligibility for a member offer does not itself authorise sending a commercial advertisement.

Tools and Data

CampaignBrief: objective, audience description, permitted channel, start/end dates, budget ceiling and restrictions.

Approved SupplierOffer and PriceBreakdown records from UC-06, with validity and source references.

LLM drafting function: brief + supplier evidence -> campaign draft, proposal explanation and source references.

Simulated channel-cost configuration and deterministic evaluation function: proposal -> total estimated cost, check results and reasons.

Reviewer/Guardrail: evidence, content and evaluation checks -> pass or revision reasons.

Operator decision form: Approve / Request revision / Reject and comments.

Local Campaign record and Agent Trace: brief, proposal, evidence, estimated cost, status, review and decision.

Optional CampaignResponse record: campaignId, sourceChannel and response content, handed to Operations.

Approved MemberOfferRule: offer ID, applicable supplier offers, eligible membership tiers, fixed discount amount, validity period and human approval record.

This specification does not require advertising APIs, payment integrations, or predictive models.

Approval conditions

Every campaign proposal requires human approval after review. A change to budget ceiling, content restrictions or permitted channel requires operator confirmation. External publication or actual spending additionally requires explicit action authorisation. The prototype ends at the approved proposal or optional simulation.

Acceptance Criteria

AC-UC08-01: Given a brief without a budget ceiling or dates, the system requests the missing information and does not submit an invented brief for approval.

AC-UC08-02: Given an expired/unapproved offer or unsupported advertised price, the campaign is blocked for revision with a reason and does not become Approved.

AC-UC08-03: Given 20 simulated campaign units at AUD 2.00 each and a fixed cost of AUD 10.00, total estimated cost is AUD 50.00. A ceiling of AUD 60.00 passes the budget check; a ceiling of AUD 45.00 fails. These figures are fixtures, not agreed channel pricing.

AC-UC08-04: Given an unsupported supplier claim or a failed content rule, Reviewer requests revision and no approval is inferred.

AC-UC08-05: Approve saves the reviewed campaign; Request revision clears approval and repeats checks; Reject records the reason; if no decision is recorded, the campaign remains in AwaitingApproval.

AC-UC08-06: A tool/save failure does not return an approved result. A successfully approved proposal includes supplier evidence, cost breakdown and the recorded human decision.

AC-UC08-07: Material changes to the proposal or selected supplier data require reassessment and fresh approval before further use.

AC-UC08-08: Approving or previewing the proposal does not publish an advertisement or spend real money. If a simulated response is produced, it includes campaign/source references and goes to Operations.

AC-UC08-09: An approved member offer records its eligibility conditions, discount, applicable SupplierOffers and validity period. UC-07 receives the eligibility rule, while UC-06 receives the approved discount rule and rechecks the selling price and margin. Neither component invents missing terms.

AC-UC08-10: A campaign targeting Paid Members with commercial advertising fails review. Member-offer eligibility does not bypass this restriction.

Assumptions and scope

This prototype uses simulated offers and channel costs, one AUD cost basis, and simple deterministic checks. Permitted channels, channel cost units, content restrictions, and campaign approval conditions are configuration values rather than fixed platform or pricing defaults in this requirement. Acceptance-criteria amounts are test fixtures only. Audience descriptions use supplied simulated or non-personal data; personalised ad targeting and live advertising-platform integration are outside the base prototype.

8 Activity Diagrams

8.1 Operations Agent

AD-01 models a customer planning request from identity verification and request confirmation through entitlement checks, specialist routing, review, approval, controlled delivery, and closure. It also shows clarification, human-assistance, retry, revision, and persistent-delivery-failure paths for UC-05 and UC-07.

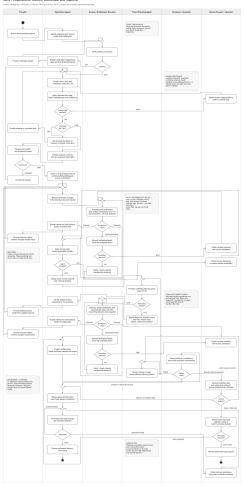


Fig. 4. AD-01 — Operations Request Intake, Coordination, and Response Activity

See page 35 for an enlarged version.

8.2 Travel Planning Agent

AD-02 covers both new-plan creation and confirmed revision. The Travel Planning Agent retrieves approved offers, constructs candidates, applies deterministic feasibility and price checks, ranks valid options, compares a revision with its baseline, and returns a reviewed result or an actionable issue to Operations.

8.3 Customer Exception Agent

AD-03 models the UC-03 disruption-recovery activity for a flight cancellation. It traces evidence collection, itinerary-impact analysis, recovery-option validation, review, Traveller selection, approval of restricted commitments, outcome delivery, and escalation when evidence, options, decisions, or delivery remain unresolved.

8.4 Supplier Partnership Agent

AD-04 shows the UC-06 assessment of a supplier offer. The Supplier Partnership Agent extracts offer data, validates fields and availability, calculates price and margin, obtains review

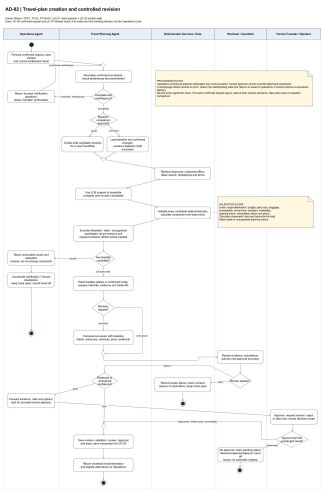


Fig. 5. AD-02 — Travel Plan Creation and Controlled Revision Activity

See page 36 for an enlarged version.

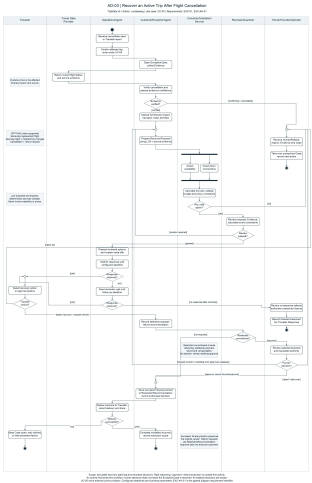


Fig. 6. AD-03 — Active-Trip Recovery After Flight Cancellation

See page 37 for an enlarged version.

and any required human decision, then records an approved, revision-required, or rejected outcome for Operations.

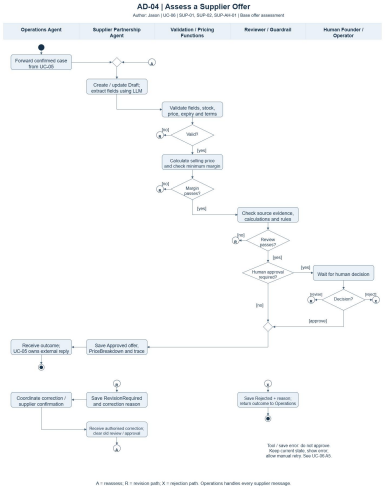


Fig. 7. AD-04 — Supplier Offer Assessment and Margin Validation Activity

See page 38 for an enlarged version.

9 Interaction Diagrams

9.1 Operations Agent

SD-01 details the UC-05 and UC-07 interactions among the Traveller, controlled interface, Operations Agent, LLM service, Entitlement Service, Travel Planning Agent, Reviewer/Guardrail, and Human Founder/Operator. It includes clarification, entitlement denial, unavailable handlers, approval, response delivery, retry, and recorded closure.

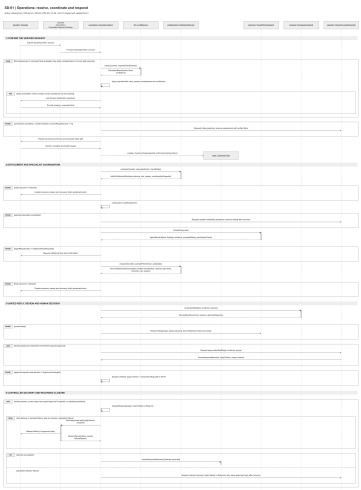


Fig. 8. SD-01 — Operations Request Coordination Sequence

See page 39 for an enlarged version.

9.2 Travel Planning Agent

SD-02 models the UC-01 creation sequence. It shows constraint extraction, clarification, approved-offer retrieval, candidate assembly, deterministic validation, ranking, review, possible human decision, and return of saved versions and evidence to Operations.

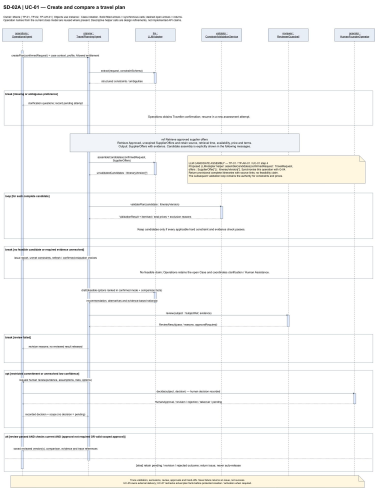


Fig. 9. SD-02 — Create and Compare a Travel Plan Sequence

See page 40 for an enlarged version.

9.3 Customer Exception Agent

SD-03 traces the UC-03 interaction from a cancellation notice or Traveller report through case intake, evidence verification, recovery assessment, reviewed options, Traveller response, restricted-commitment approval, persistence, and release through Operations. It also models no-response, rejection, escalation, and unresolved-closure paths.

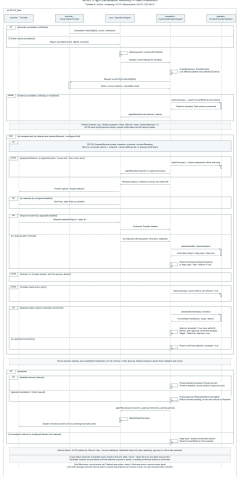


Fig. 10. SD-03 — Flight Cancellation Recovery Interaction

See page 41 for an enlarged version.

9.4 Supplier Partnership Agent

SD-04 shows the UC-06 sequence among Operations, the Supplier Partnership Agent, LLM extraction, deterministic validation and pricing, Reviewer/Guardrail, and the Human Founder/Operator. Failed checks request correction; successful assessment records the approved offer and PriceBreakdown before Operations handles the external response.

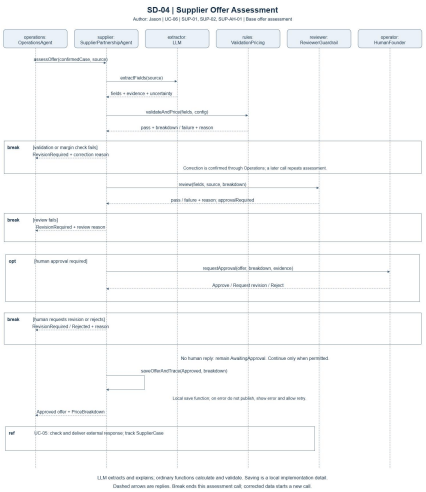


Fig. 11. SD-04 — Supplier Offer Assessment Sequence

See page 42 for an enlarged version.

10 State Machine Diagrams

10.1 Operations Agent

SM-01 defines the lifecycle of an Operations-managed CustomerCase, from confirmation and assignment through specialist work, access decisions, review, approval, response readiness, delivery, and closure. Waiting, verification failure, revision, human assistance, withdrawal, retry, and authorised manual closure remain explicit states or transitions.

10.2 Travel Planning Agent

SM-02 models an ItineraryVersion assessment from Draft through validation, feasibility, review, possible human review, and hand-off. Infeasible, revision-required, rejected, stale-evidence, and material-change paths preserve history and prevent release until current checks and any scoped approval are valid.

10.3 Customer Exception Agent

SM-03 defines the UC-03 ExceptionCase lifecycle for disruption recovery. It covers detection, investigation, recovery

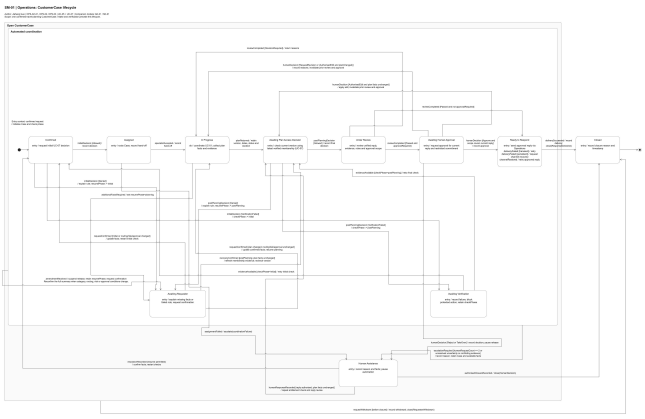


Fig. 12. SM-01 — Operations CustomerCase Lifecycle

See page 43 for an enlarged version.

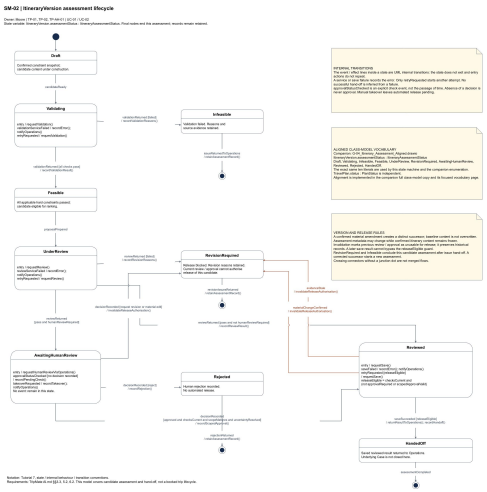


Fig. 13. SM-02 — ItineraryVersion Assessment Lifecycle

See page 44 for an enlarged version.

planning, review, Traveller and human decisions, response delivery, external action, resolution, unresolved closure, and human takeover while preserving evidence and preventing an unverified outcome from being treated as complete.

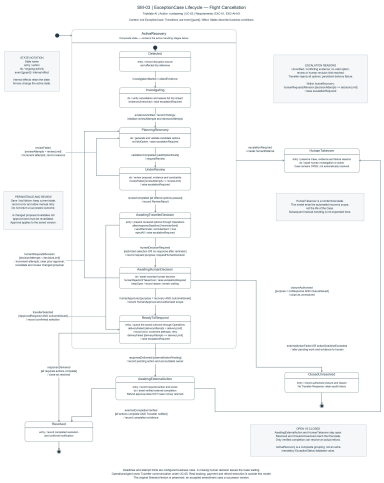


Fig. 14. SM-03 — ExceptionCase Lifecycle for Flight Cancellation Recovery

See page 45 for an enlarged version.

10.4 Supplier Partnership Agent

SM-04 models a SupplierOffer from Draft through validation, review, approval, revision, rejection, or expiry. Only approved and unexpired offers are selectable for planning; corrections clear prior decisions, and validation, review, human-decision, or save failures cannot silently produce an approved offer.

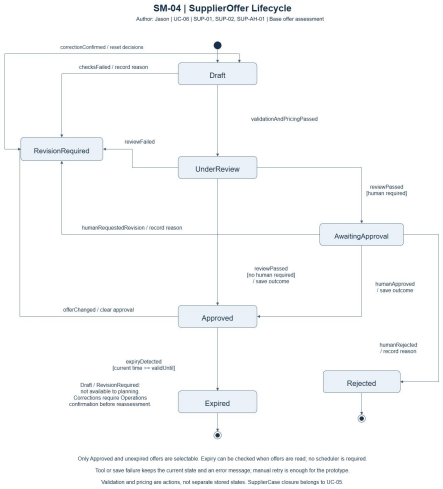


Fig. 15. SM-04 — SupplierOffer Assessment Lifecycle

See page 46 for an enlarged version.

11 Elementary Structure Diagram

G-04 presents the system-wide class model for actors, agents, cases, travel plans, itinerary versions, supplier offers, campaign records, evidence, decisions, services, and traceability. Associations and interfaces provide the shared structural vocabulary reused by the behavioural and complex-structure diagrams.



Fig. 16. G-04 — TripMate AI Elementary Structure Model

See page 47 for an enlarged version.

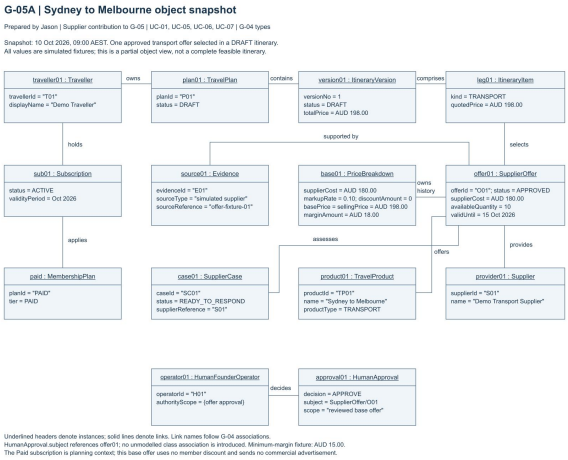


Fig. 17. G-05A — Supplier Offer Assessment Object Snapshot

See page 48 for an enlarged version.

12 Complex Structure Diagram

G-05A, G-05B and G-05C describe the Supplier Partnership contribution to the group runtime model. The scenario uses a simulated Sydney-to-Melbourne transport offer and the existing G-04 classifiers.

G-05A — Object Diagram. G-05A is an object snapshot linking a Traveller and membership context to a draft itinerary item, an approved SupplierOffer, its source Evidence and PriceBreakdown. The base cost is AUD 180.00, markup is 10%, selling price is AUD 198.00 and margin is AUD 18.00. The minimum-margin fixture is AUD 15.00. The draft contains only the illustrated transport item and is not represented as a complete feasible itinerary.

G-05B — Communication Diagram. G-05B models the successful UC-06 supplier-offer assessment scenario requiring human approval. Numbered messages show Operations requesting the assessment, followed by LLM extraction, deterministic validation and pricing, Reviewer/Guardrail review and the Human Founder/Operator decision. Failed checks and revision behaviour remain specified in UC-06 and SD-04. Operations retains responsibility for external communication.

G-05C — Composite Structure Diagram. G-05C models the internal runtime structure of SupplierAssessmentContext. Its typed parts reuse SupplierPartnershipAgent, LLMAdapter,

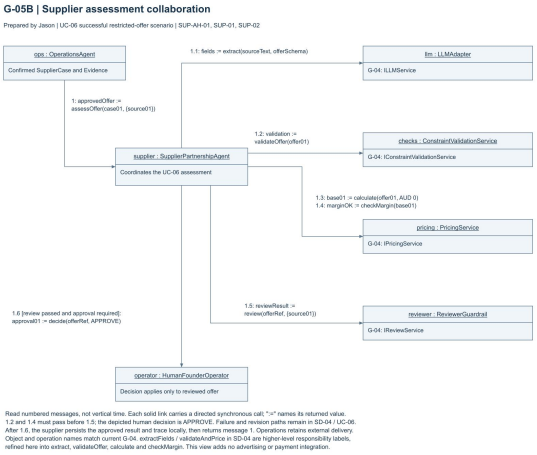


Fig. 18. G-05B — Supplier Offer Assessment Communication Diagram

See page 49 for an enlarged version.

ConstraintValidationService, PricingService and ReviewerGuardrail. Ports and connectors represent service contracts and delegation of intake and human-decision requests. SupplierAssessmentContext is a runtime design refinement rather than an additional business-domain entity or deployment unit.

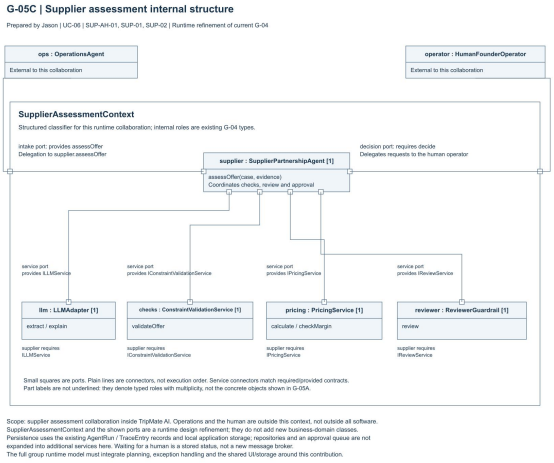


Fig. 19. G-05C — SupplierAssessmentContext Composite Structure

See page 50 for an enlarged version.

Together, the three views describe the Supplier Partnership runtime collaboration. Local records and ordinary function calls are sufficient for the teaching prototype; no external advertising system, payment system, background expiry service or message broker is required.

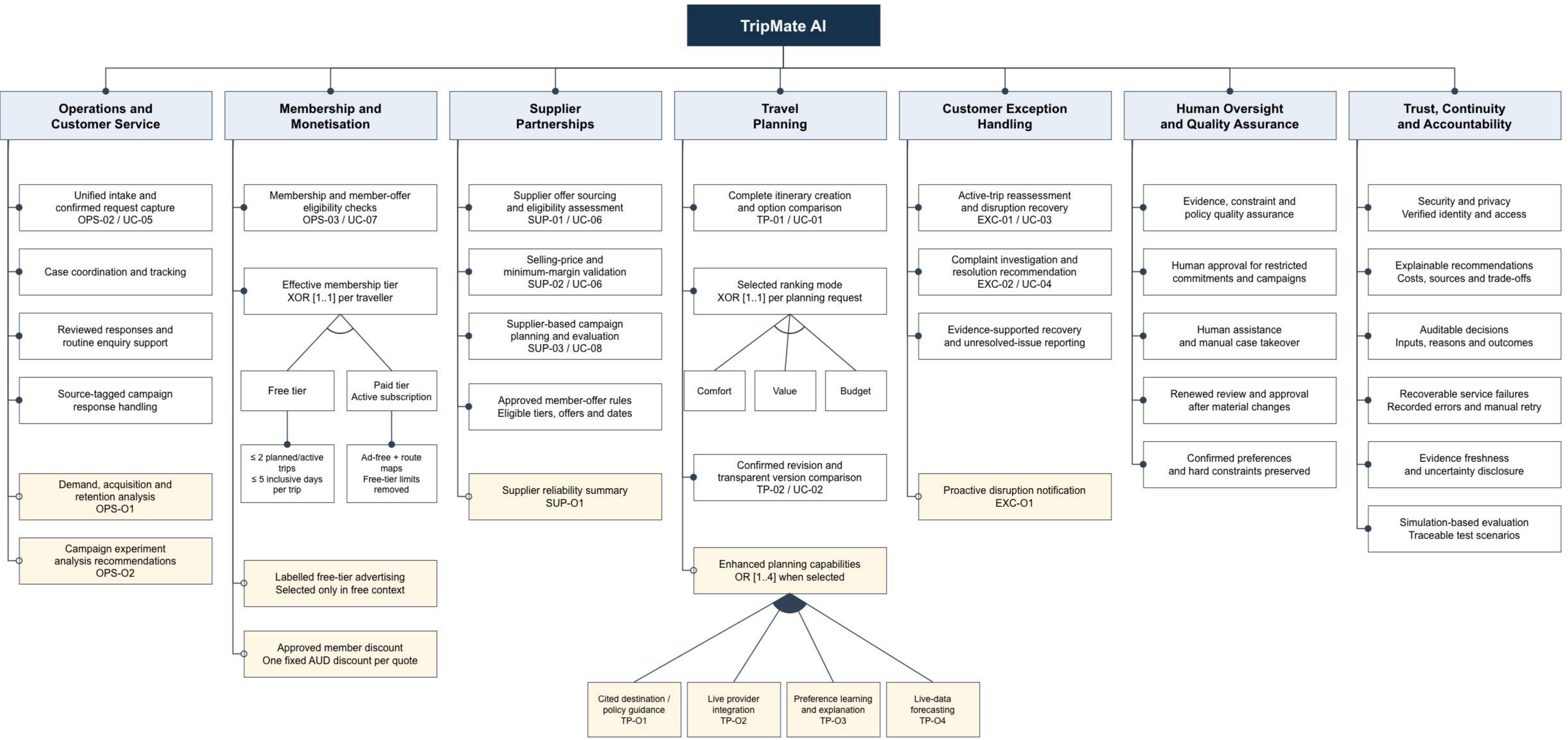
13 Assumptions

- TripMate AI is an educational prototype that uses simulated traveller, supplier, booking, payment, disruption, weather, and campaign data.
- All prototype prices and calculations use AUD, a consistent quoted quantity, and two-decimal rounding.
- Travellers and Travel Service Providers communicate through the Operations Agent rather than directly with specialist Agents.
- LLMs may interpret requests, organise information, and draft explanations, while deterministic services remain responsible for validation, entitlement decisions, pricing, and calculations.
- The Human Founder/Operator remains responsible for financial, contractual, legal, safety, publication, and other restricted decisions.
- Real bookings, payments, refunds, contracts, advertising publication, and live external-service integrations are outside the prototype scope.
- Membership limits, margin thresholds, approval conditions, and other business rules are treated as configured system values unless explicitly stated.

14 Appendix

G-02 | TripMate AI — Feature Model

Scope: mandatory capabilities belong to the product. XOR subgroups bind separately to one verified traveller and one planning request.
These scoped choices do not remove support for other tiers / modes from the deployment; this is not a flat software-product-line configuration.

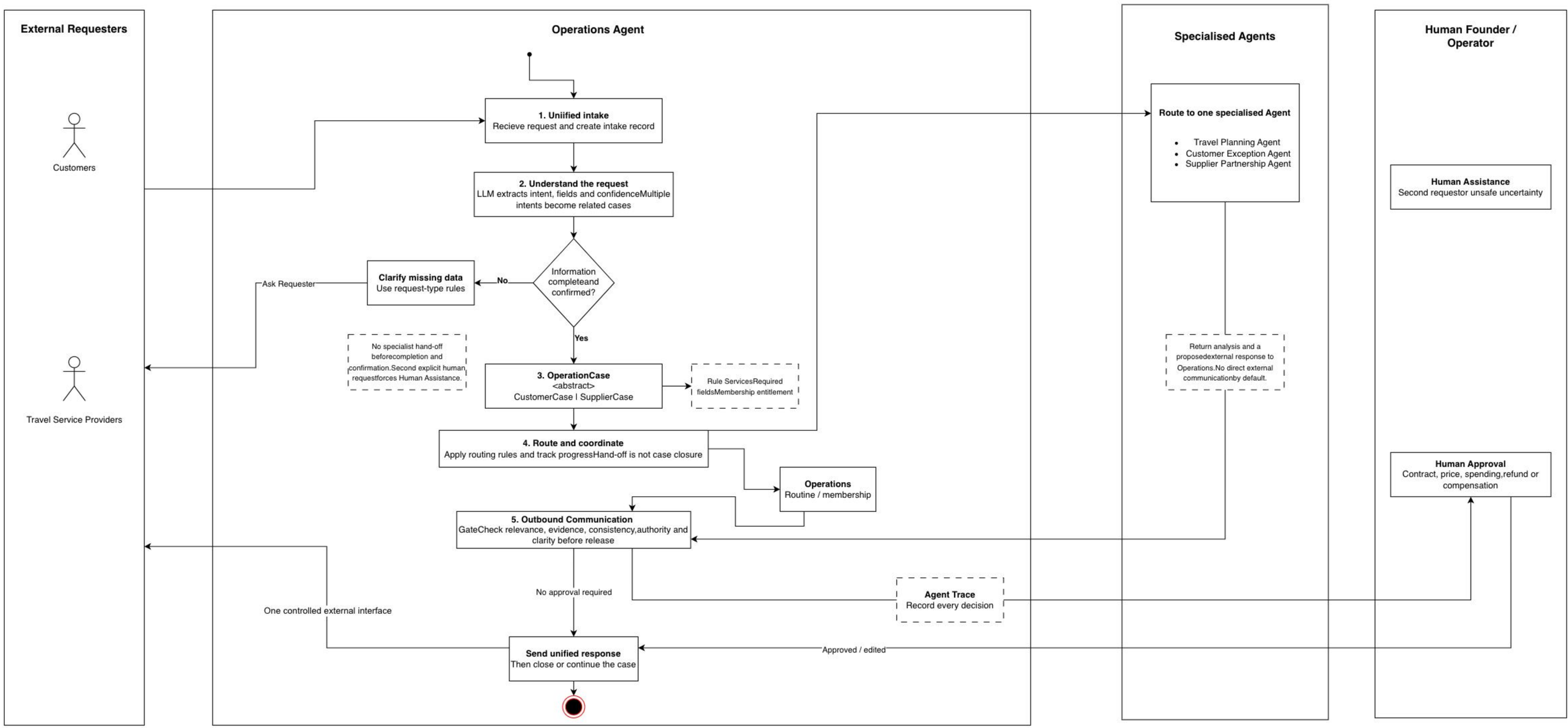


NOTATION
● Mandatory: child required if parent selected. ○ Optional: child may be selected.
Empty group arc = XOR, exactly one. Filled group arc = OR, at least one. Group-member edges have no individual mandatory circles.
Dashed requires / excludes connectors are on page 2; endpoints reference the SAME Feature IDs as this tree.

MODEL RULES
C1 PAID excludes ADS. C2 ADS requires FREE. C3 RULE requires ELIG. C4 DISCOUNT requires RULE. C5 DISCOUNT requires ELIG. C6 DISCOUNT requires PRICE.
C7 SUP-03 requires APPROVAL. C8 TP-O2 requires FRESH. C9 TP-O2 requires SECURITY. C10 TP-O3 requires CONSENT. C11 TP-O4 requires FRESH.
C1–C2 apply to the same traveller context; C4–C6 to the same quotation. These are selection constraints, not activity-flow arrows.

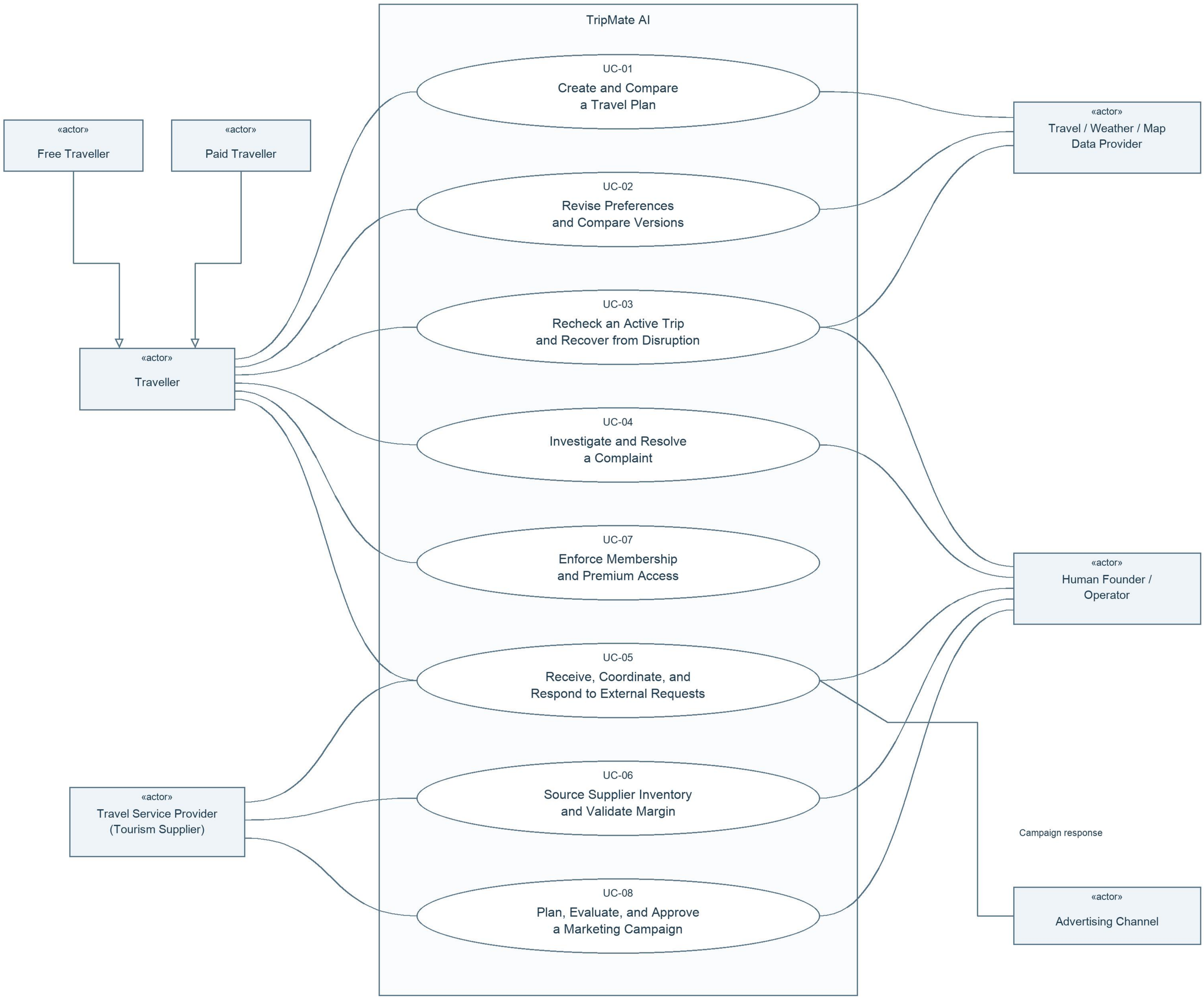
SCOPE AND TRACEABILITY
Seven mandatory domains express required product capabilities, not mandatory execution in every request. Feature presence does not cause an approval, advertisement or discount to execute.
Runtime ADVERTISING and DISCOUNT selections are contextual; the product retains the capability. Core quality / oversight controls remain mandatory.
The optional EXT parent plus its OR group permits 0–4 planning extensions without making any optional requirement mandatory. Simulated updates suffice for EXC-O1.
Sources: TripMate AI.md §§2–4; Supplier_Requirements.md; Stage1_Modeling_Task_List.md §8. Notation: ELEC5620-TUT-03, pp.12–14. TP-* follows detailed requirements.

Ad-hoc Diagram: Operations Agent



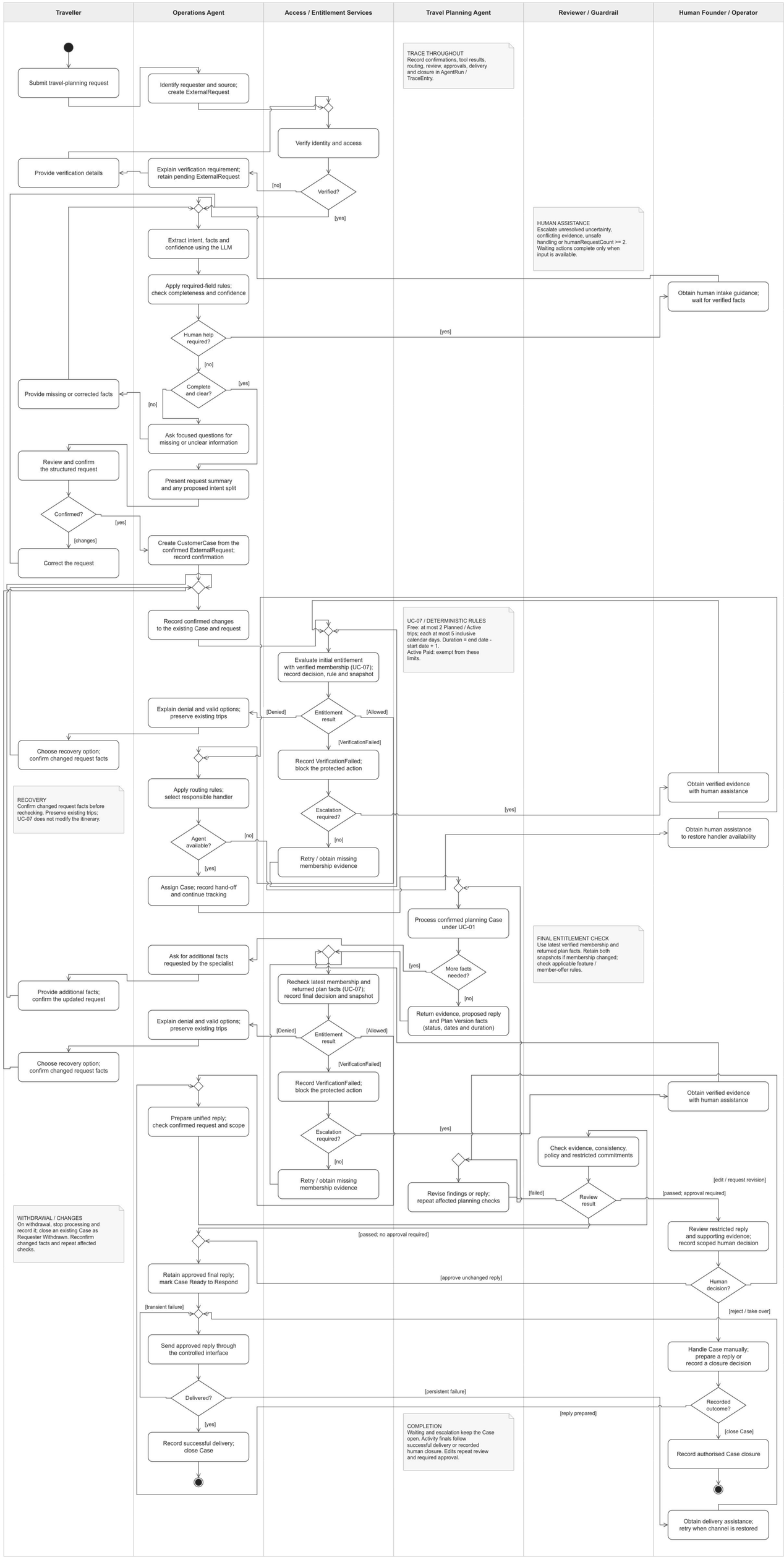
G-03 | Overall Use Case Diagram

TripMate AI · Group 6



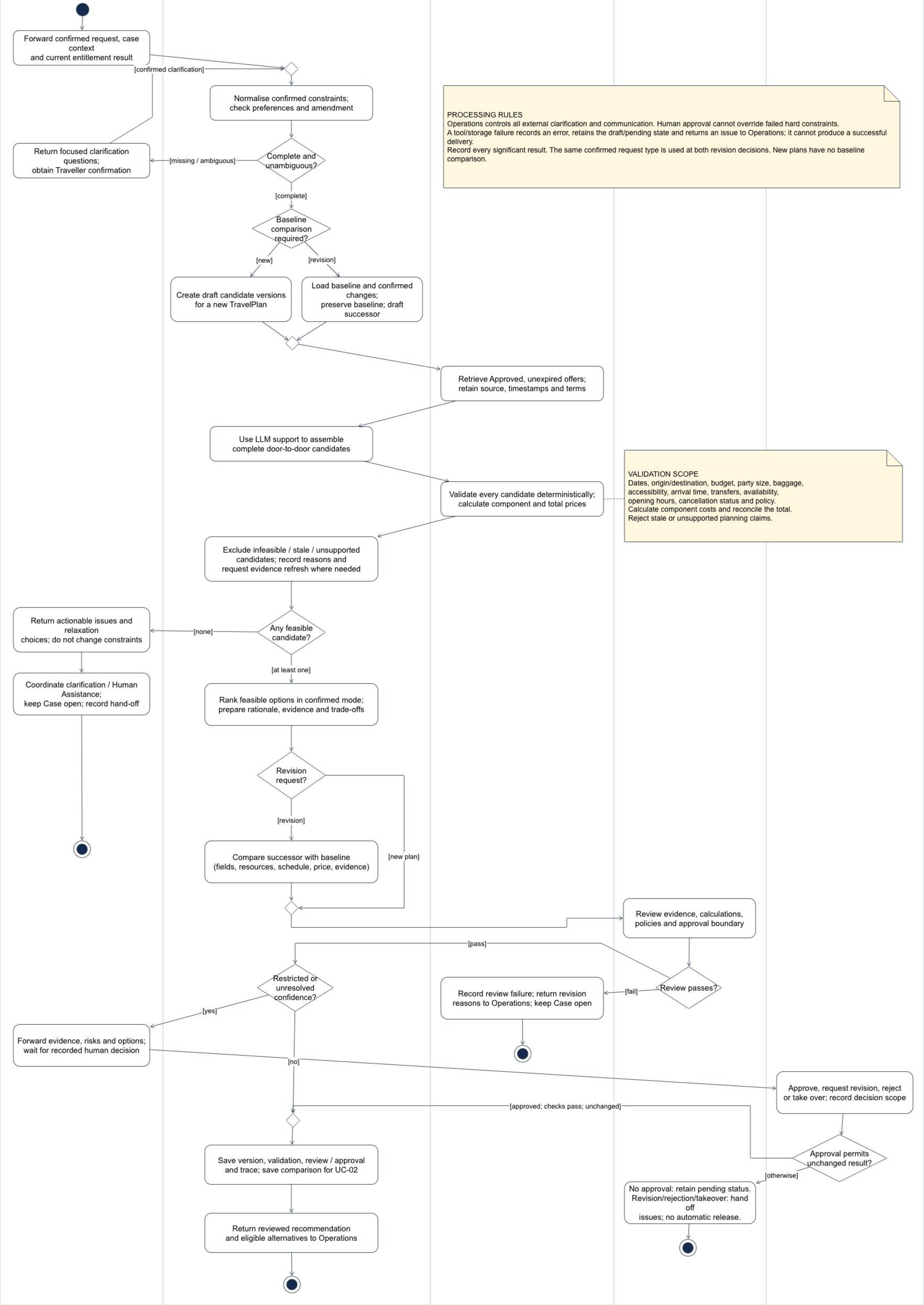
AD-01 | Operations: receive, coordinate and respond

Author: Jiaheng Guo | OPS-AH-01, OPS-02, OPS-03 | UC-05 + UC-07 | Scope: one customer travel-planning Case



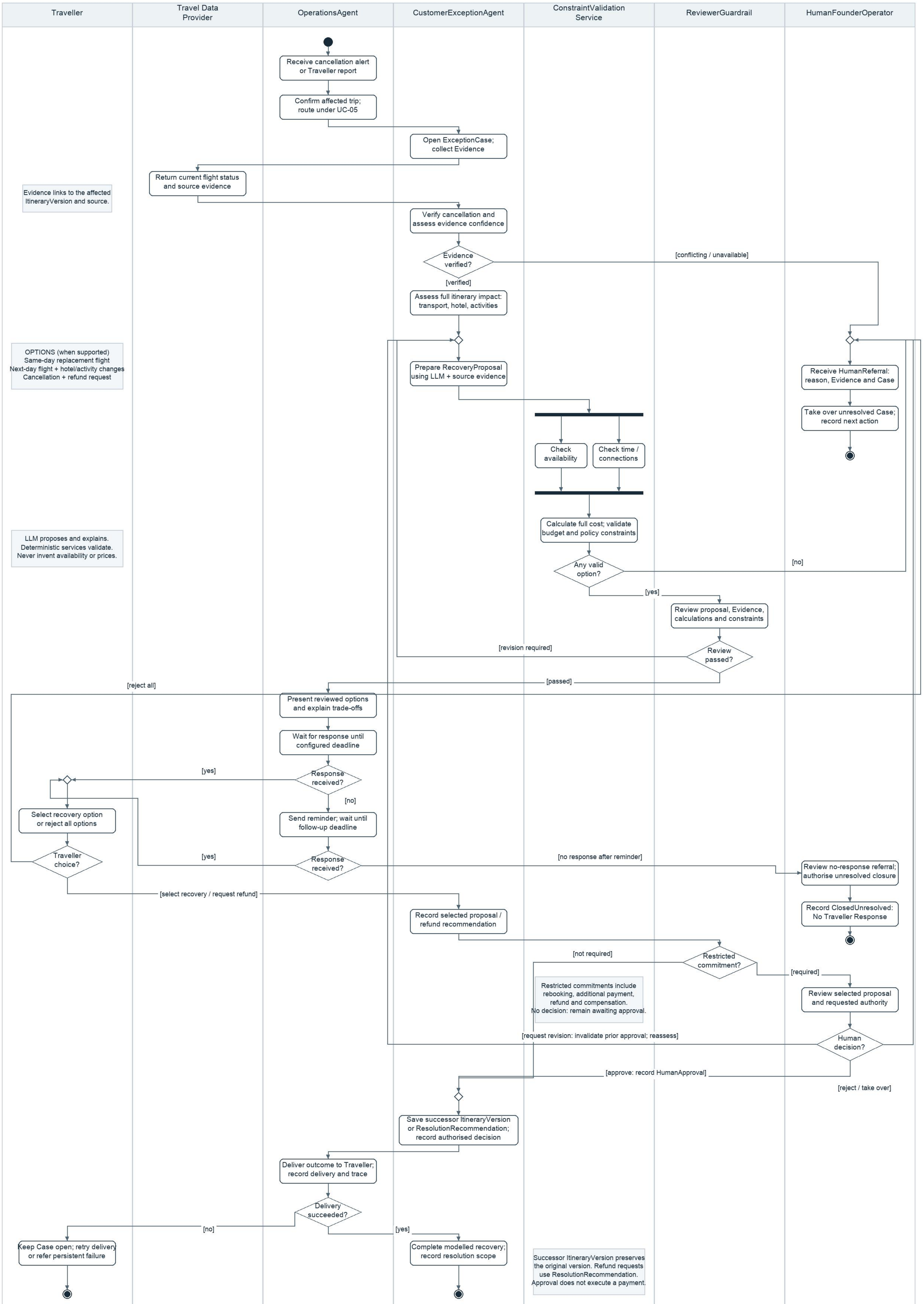
AD-02 | Travel-plan creation and controlled revision

Owner: Moore | TP-01, TP-02, TP-AH-01 | UC-01 main scenario + UC-02 revision path
Entry: UC-05 confirmed request and UC-07 Allowed result. End nodes end this handling attempt, not the Operations Case.



AD-03 | Recover an Active Trip After Flight Cancellation

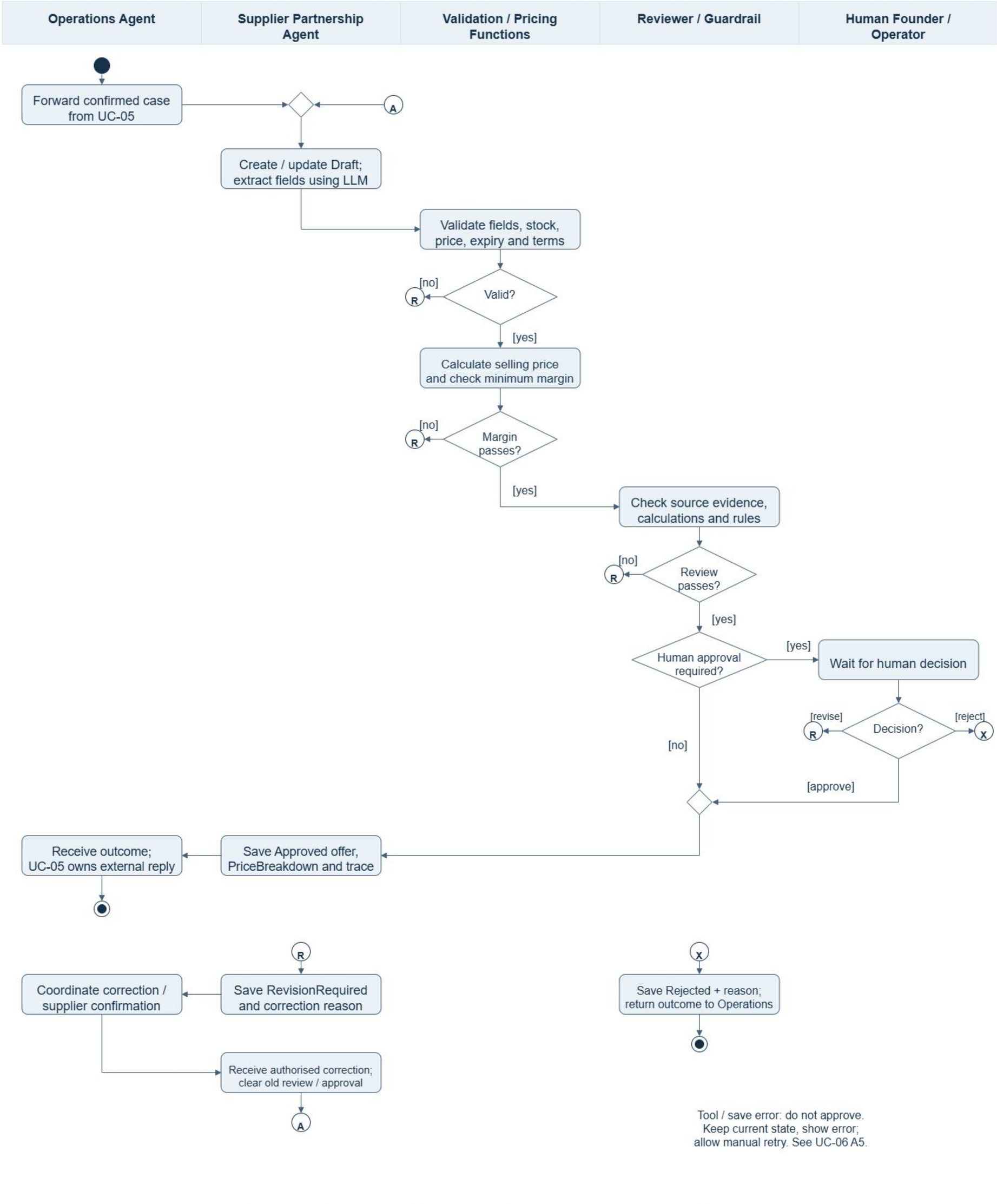
TripMate AI | Author: rundaweng | Use case: UC-03 | Requirements: EXC-01, EXC-AH-01



Scope: simulated recovery planning and recorded decisions. Real rebooking / payment / refund execution is outside this activity.
An activity final ends this workflow; human takeover does not mean the ExceptionCase is resolved. All material decisions are traced.
UC-05 owns external communication. Configured deadlines are business parameters; EXC-AH-01 is the agreed diagram requirement identifier.

AD-04 | Assess a Supplier Offer

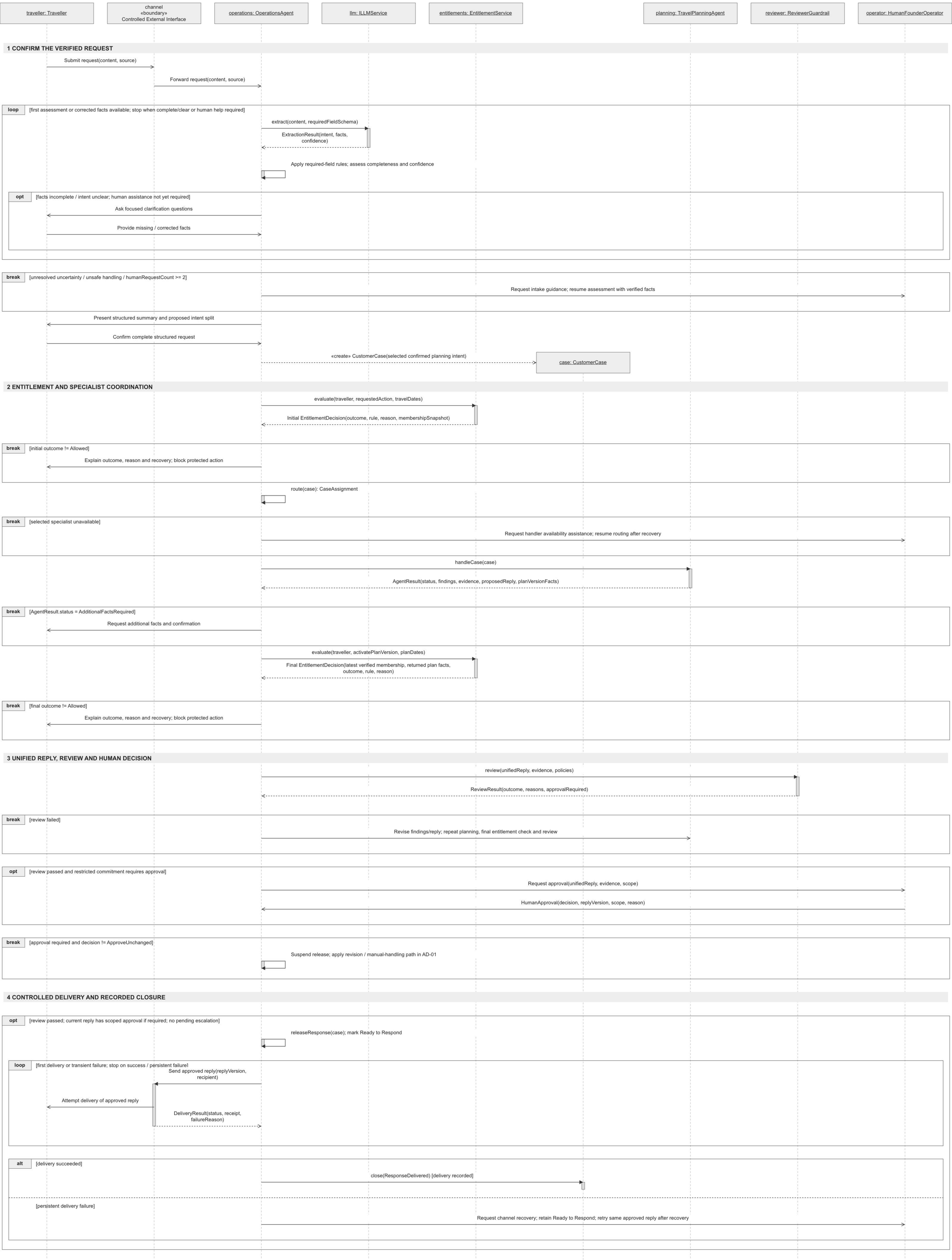
Author: Jason | UC-06 | SUP-01, SUP-02, SUP-AH-01 | Base offer assessment



A = reassess; R = revision path; X = rejection path. Operations handles every supplier message.

SD-01 | Operations: receive, coordinate and respond

Author: Jiaheng Guo | OPS-AH-01, OPS-02, OPS-03 | UC-05 + UC-07 | Aligned with updated AD-01



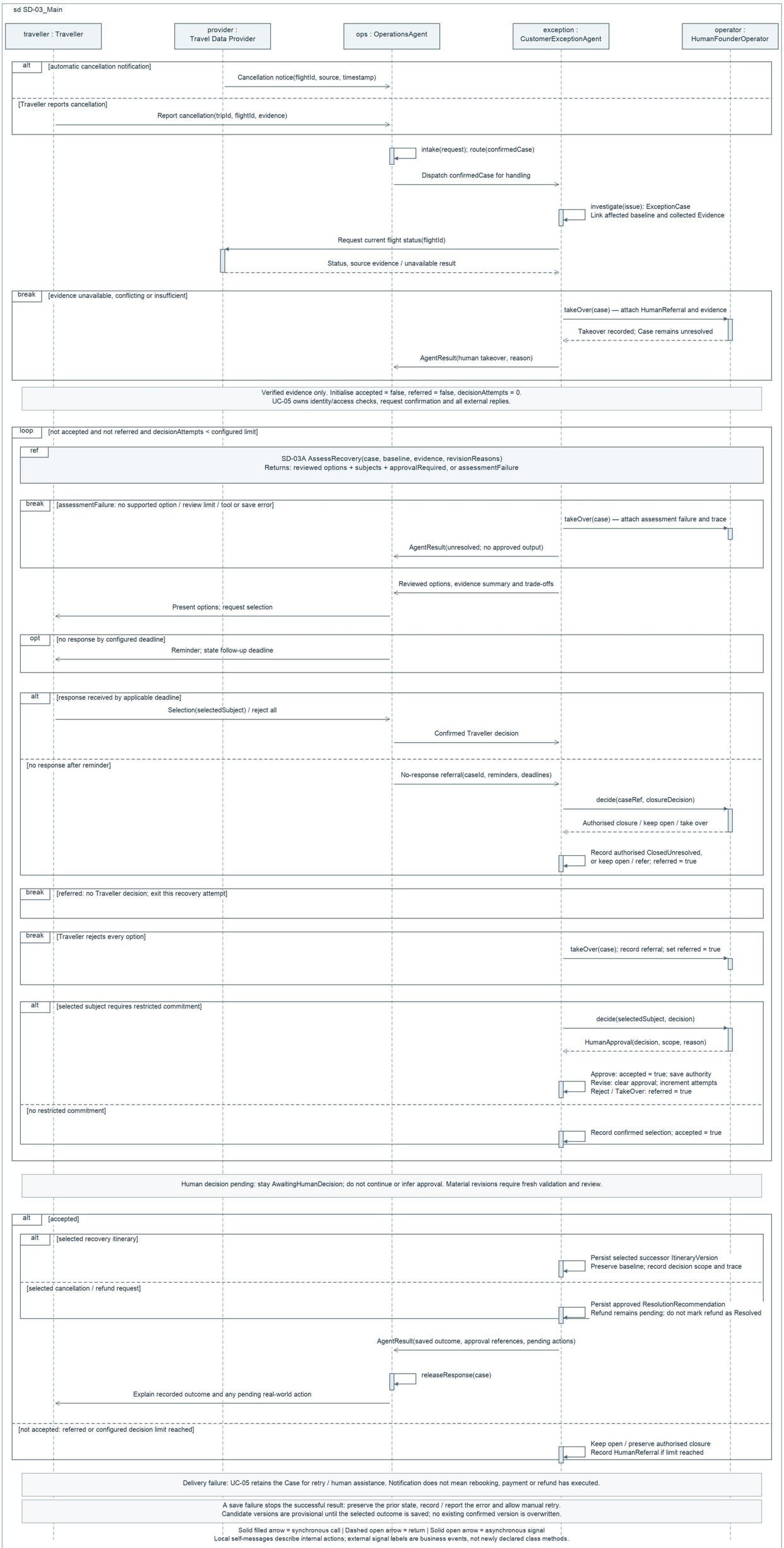
SD-02A | UC-01 — Create and compare a travel plan

Owner: Moore | TP-01, TP-02, TP-AH-01 | Objects use instance : Class notation. Solid filled arrows = synchronous calls; dashed open arrows = returns.
Operation names from the current class model are reused where present. Descriptive helper calls are design refinements, not implemented API claims.



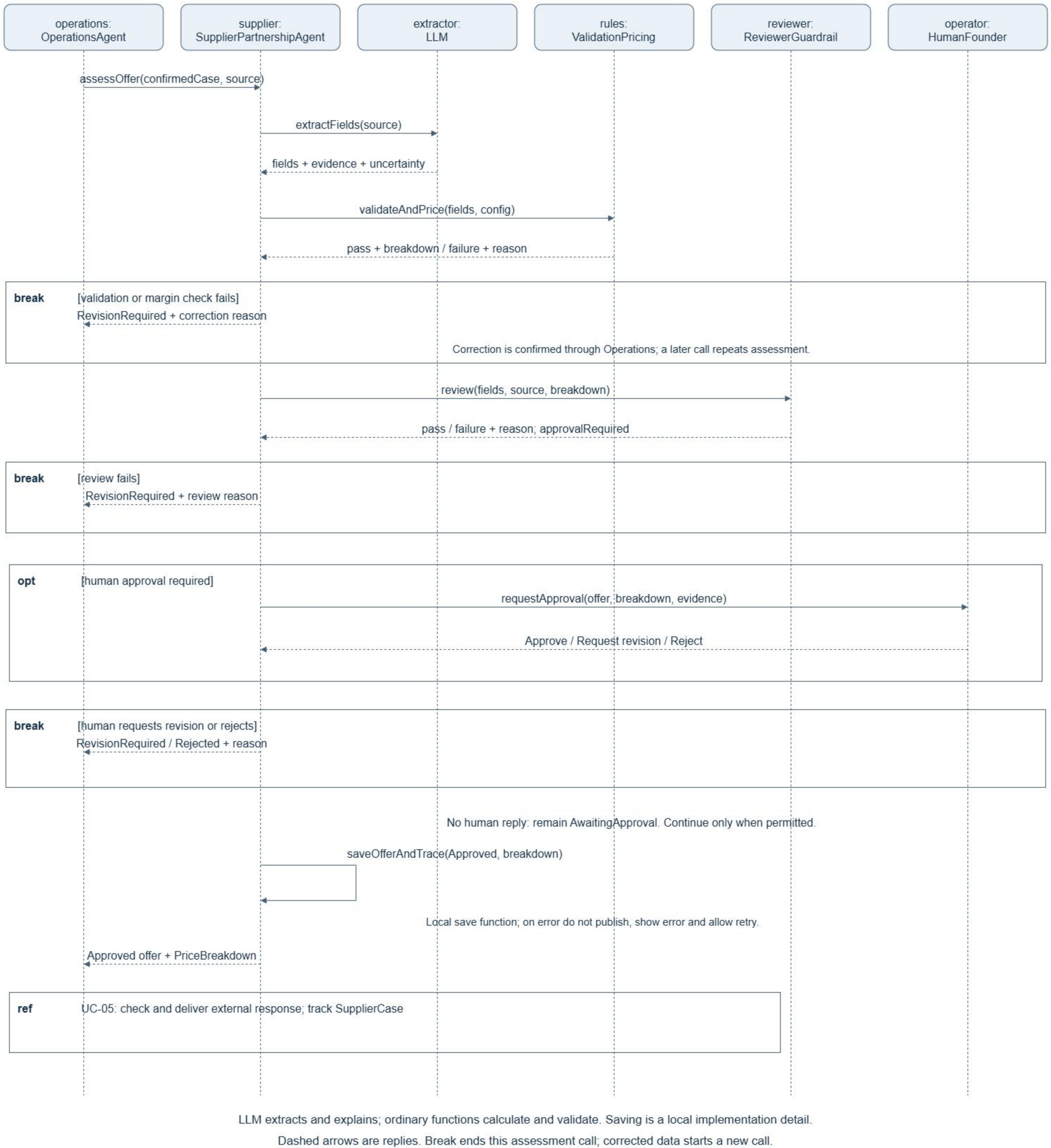
SD-03 | Flight Cancellation Recovery — Main Interaction

TripMate AI | Author: rundaweng | UC-03 | Requirements: EXC-01, EXC-AH-01



SD-04 | Supplier Offer Assessment

Author: Jason | UC-06 | SUP-01, SUP-02, SUP-AH-01 | Base offer assessment

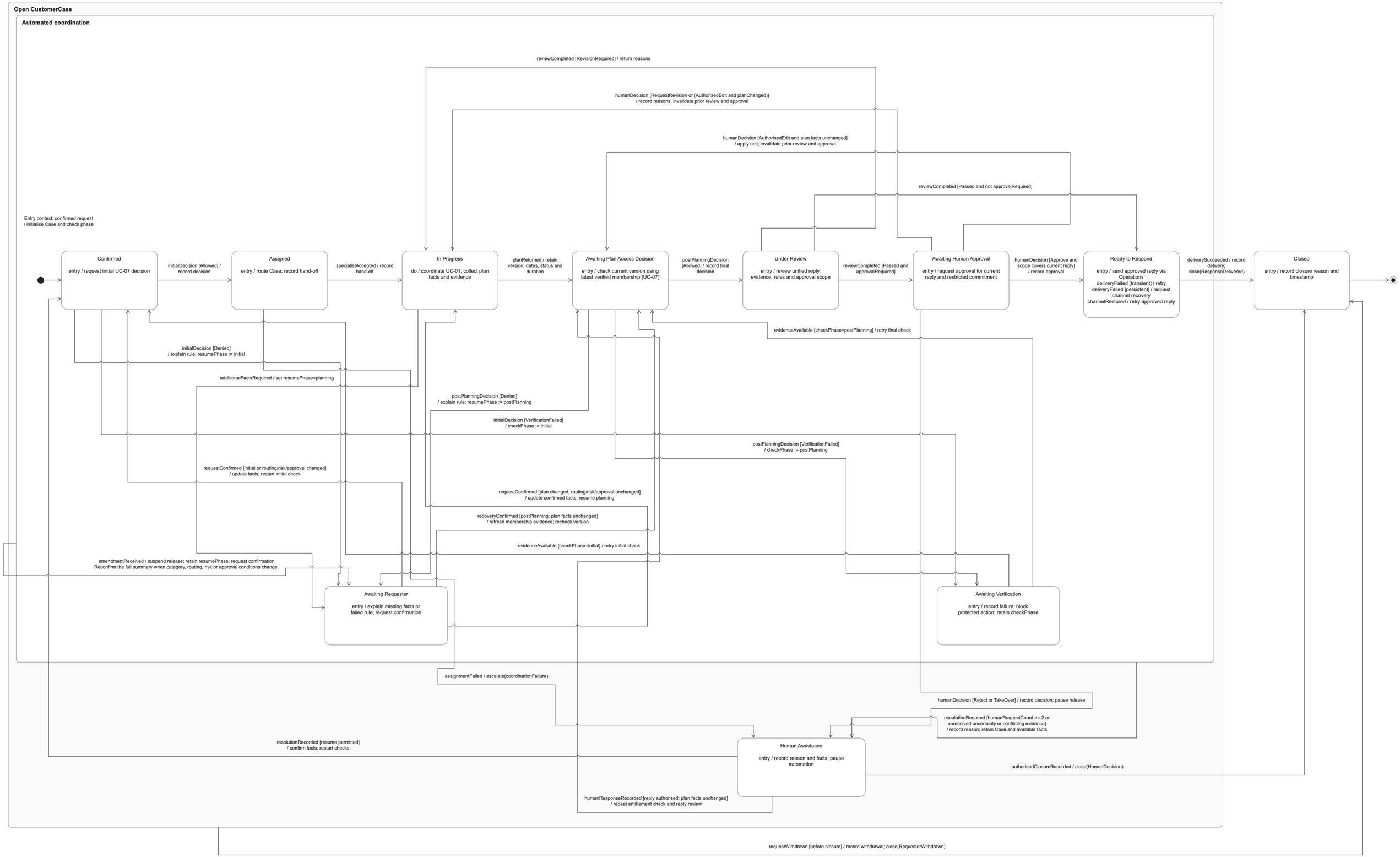


LLM extracts and explains; ordinary functions calculate and validate. Saving is a local implementation detail.

Dashed arrows are replies. Break ends this assessment call; corrected data starts a new call.

SM-01 | Operations: CustomerCase lifecycle

Author: Jiaheng Guo | OPS-AH-01, OPS-02, OPS-03 | UC-05 + UC-07 | Companion models: AD-01 / SD-01
Scope: one confirmed travel-planning CustomerCase. Intake and verification precede this lifecycle.



SM-02 | ItineraryVersion assessment lifecycle

Owner: Moore | TP-01, TP-02, TP-AH-01 | UC-01 / UC-02
State variable: ItineraryVersion.assessmentStatus : ItineraryAssessmentStatus. Final nodes end this assessment; records remain retained.

INTERNAL TRANSITIONS
The event / effect lines inside a state are UML internal transitions: the state does not exit and entry actions do not repeat.
A service or save failure records the error. Only retryRequested starts another attempt. No successful hand-off is inferred from a failure.
approvalStatusChecked is an explicit check event, not the passage of time. Absence of a decision is never approval. Manual takeover leaves automated release pending.

ALIGNED CLASS-MODEL VOCABULARY
Companion: G-04_Itinerary_Assessment_Aligned.drawio
ItineraryVersion.assessmentStatus : ItineraryAssessmentStatus
Draft, Validating, Infeasible, Feasible, UnderReview, RevisionRequired, AwaitingHumanReview, Reviewed, Rejected, HandedOff.
The exact same ten literals are used by this state machine and the companion enumeration.
TravelPlan.status : PlanStatus is independent.
Alignment is implemented in the companion full class-model copy and its focused vocabulary page.

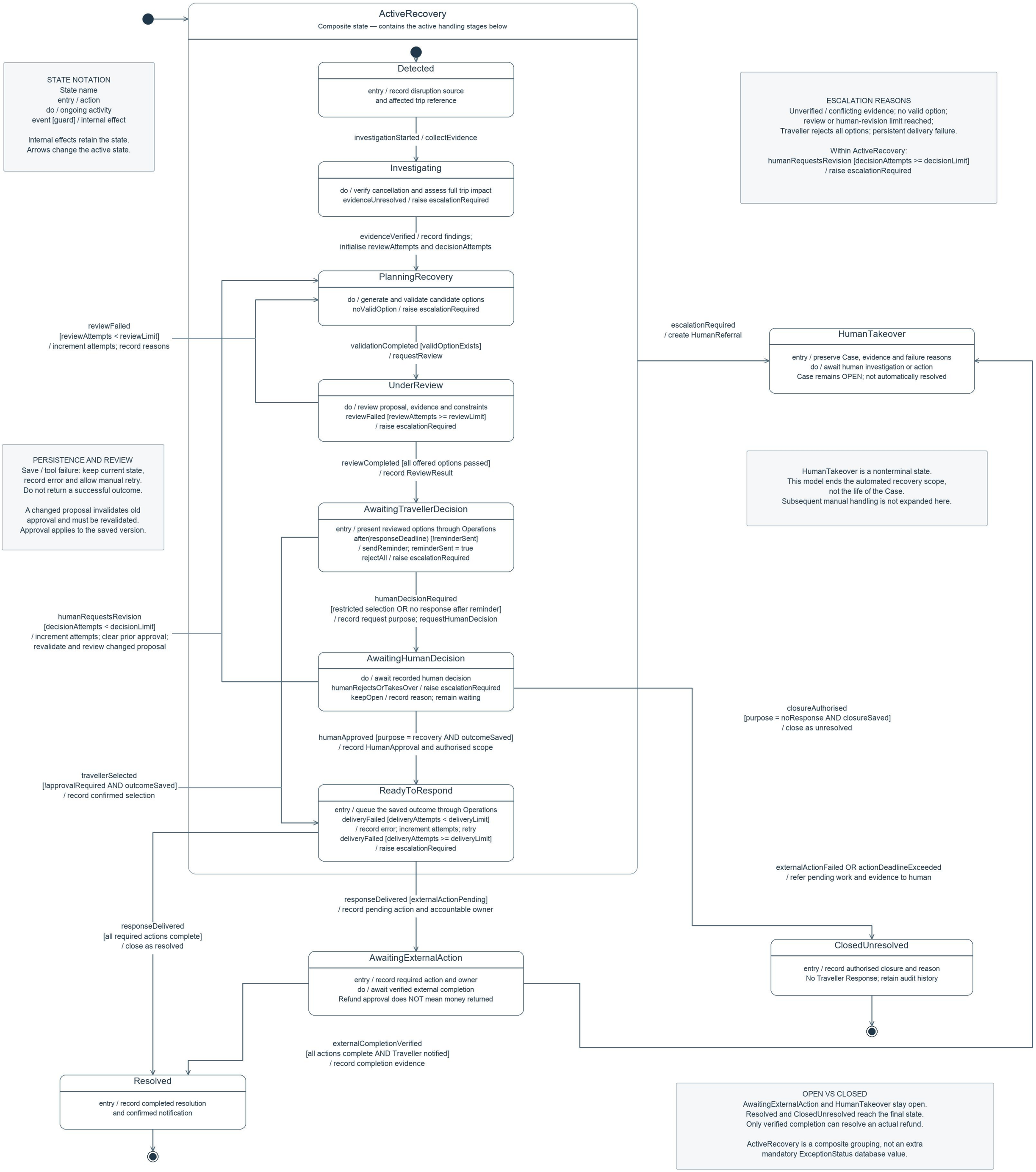
VERSION AND RELEASE RULES
A confirmed material amendment creates a distinct successor; baseline content is not overwritten.
Assessment metadata may change while confirmed itinerary content remains frozen.
Invalidation marks previous review / approval as unusable for release; it preserves historical records. A later save result cannot bypass the releaseEligible guard.
RevisionRequired and Infeasible conclude this candidate assessment after issue hand-off. A corrected successor starts a new assessment.
Crossing connectors without a junction dot are not merged flows.

Notation: Tutorial 7, state / internal behaviour / transition conventions.
Requirements: TripMate AI.md §§3.3, 5.2, 6.2. This model covers candidate assessment and hand-off, not a booked trip lifecycle.

SM-03 | ExceptionCase Lifecycle — Flight Cancellation

TripMate AI | Author: rundaweng | UC-03 | Requirements: EXC-01, EXC-AH-01

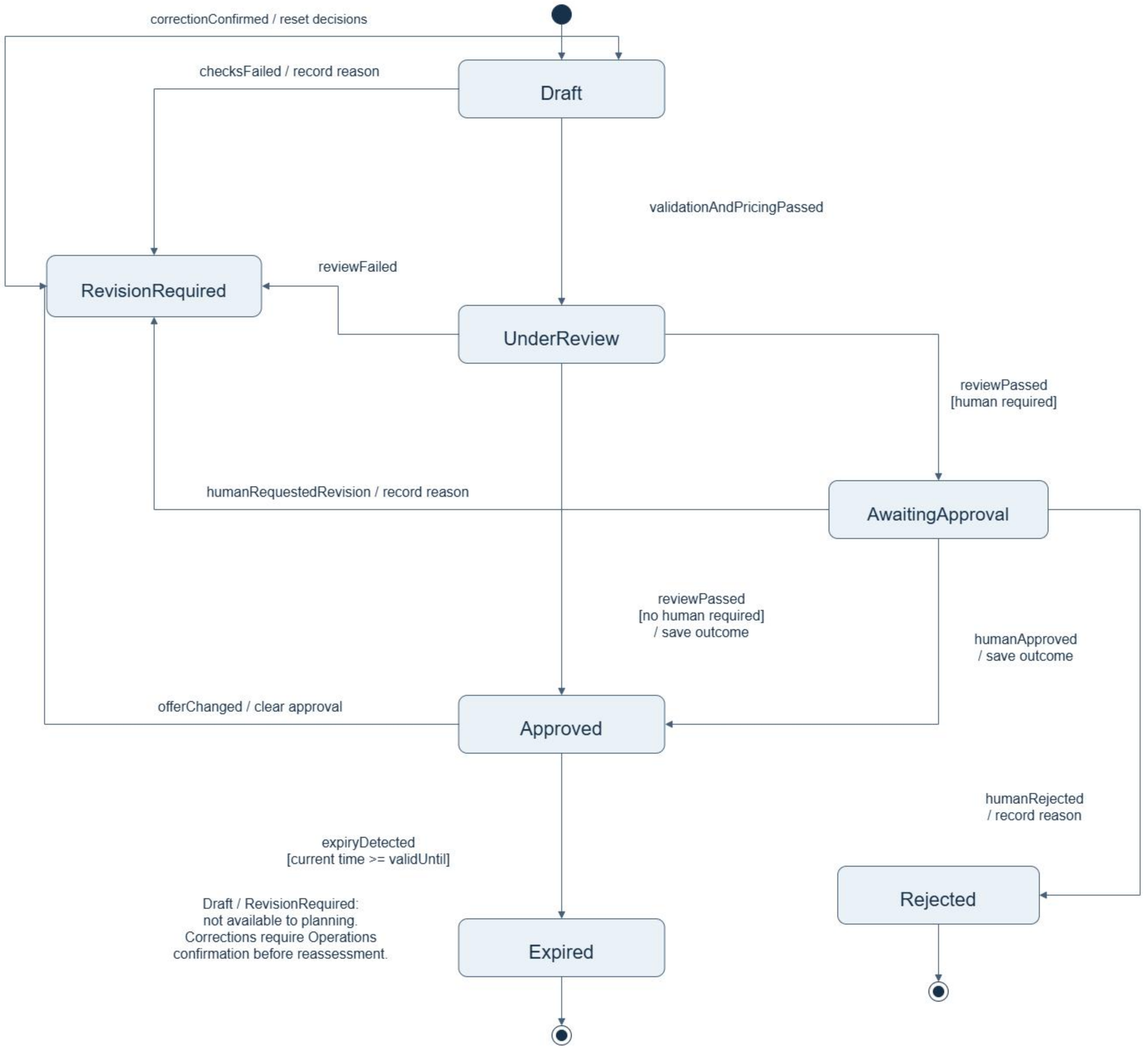
Context: one ExceptionCase. Transitions use event [guard] / effect. States describe business conditions.



Deadlines and attempt limits are configured business rules. A missing human decision leaves the Case waiting.
OperationsAgent owns Traveller communication under UC-05. Real booking, payment and refund execution is outside this model.
The original ItineraryVersion is preserved; an accepted amendment uses a successor version.

SM-04 | SupplierOffer Lifecycle

Author: Jason | UC-06 | SUP-01, SUP-02, SUP-AH-01 | Base offer assessment

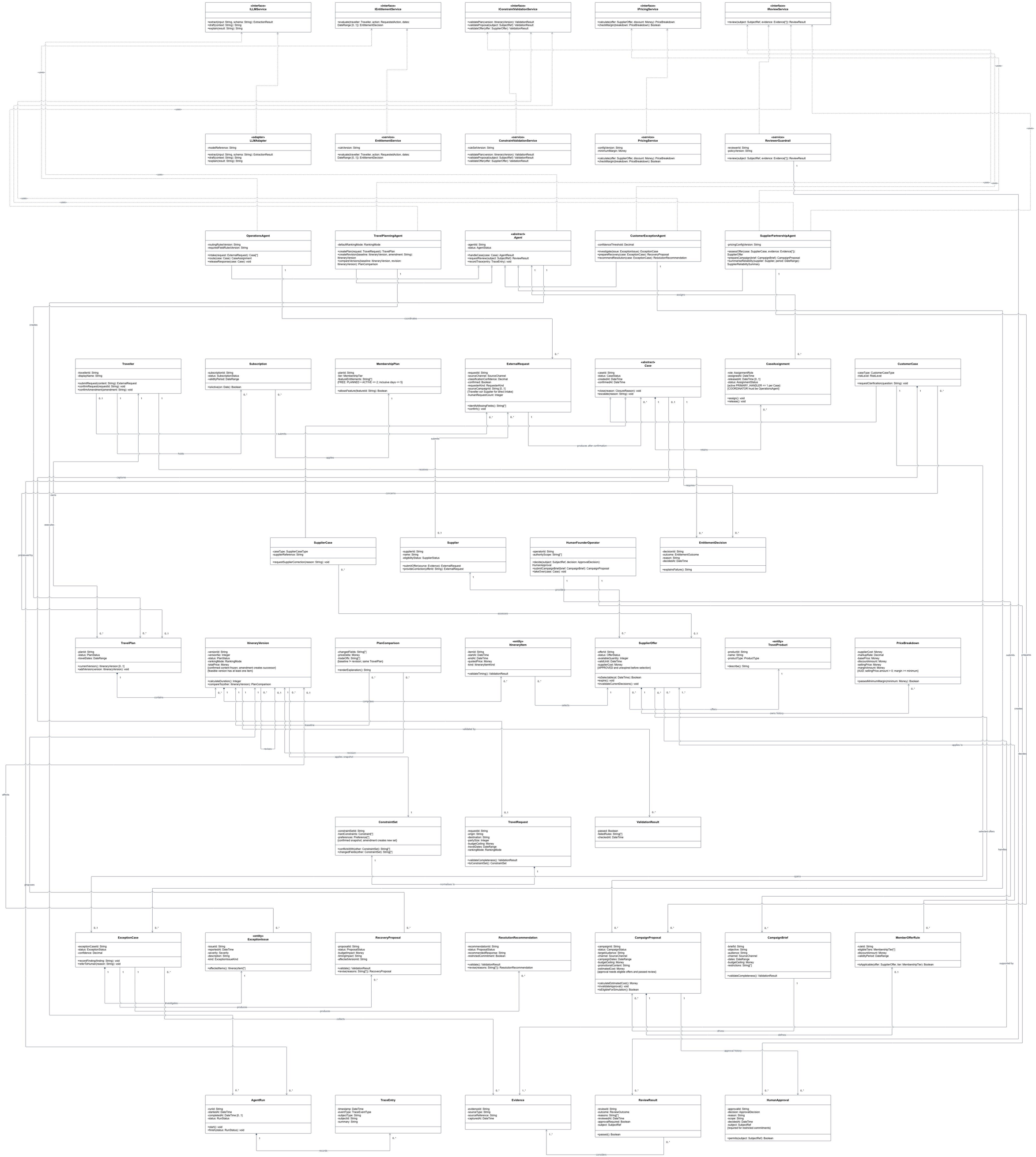


Only Approved and unexpired offers are selectable. Expiry can be checked when offers are read; no scheduler is required.

Tool or save failure keeps the current state and an error message; manual retry is enough for the prototype.

Validation and pricing are actions, not separate stored states. SupplierCase closure belongs to UC-05.

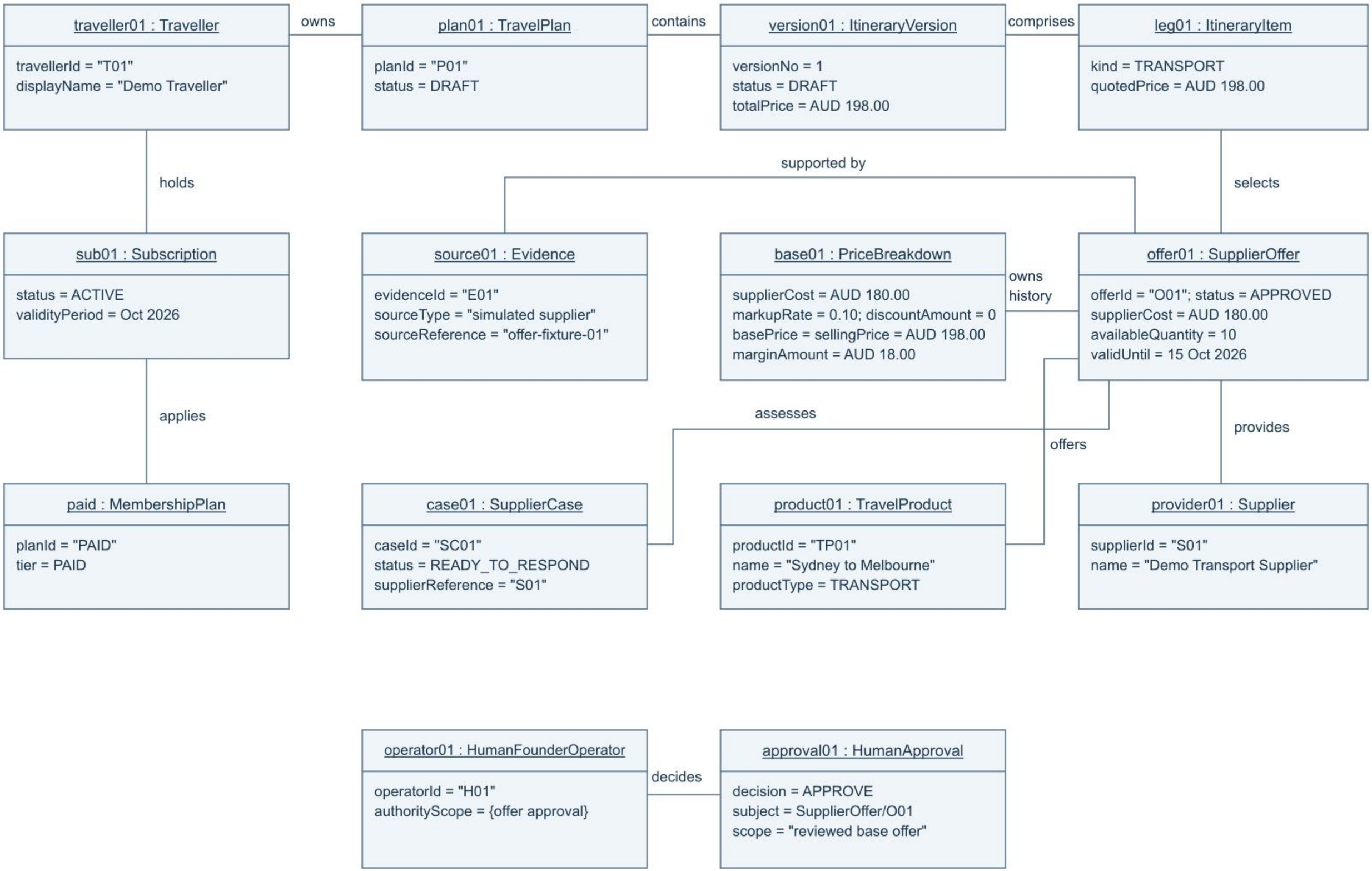
TripMate AI - Elementary Structure Model



G-05A | Sydney to Melbourne object snapshot

Prepared by Jason | Supplier contribution to G-05 | UC-01, UC-05, UC-06, UC-07 | G-04 types

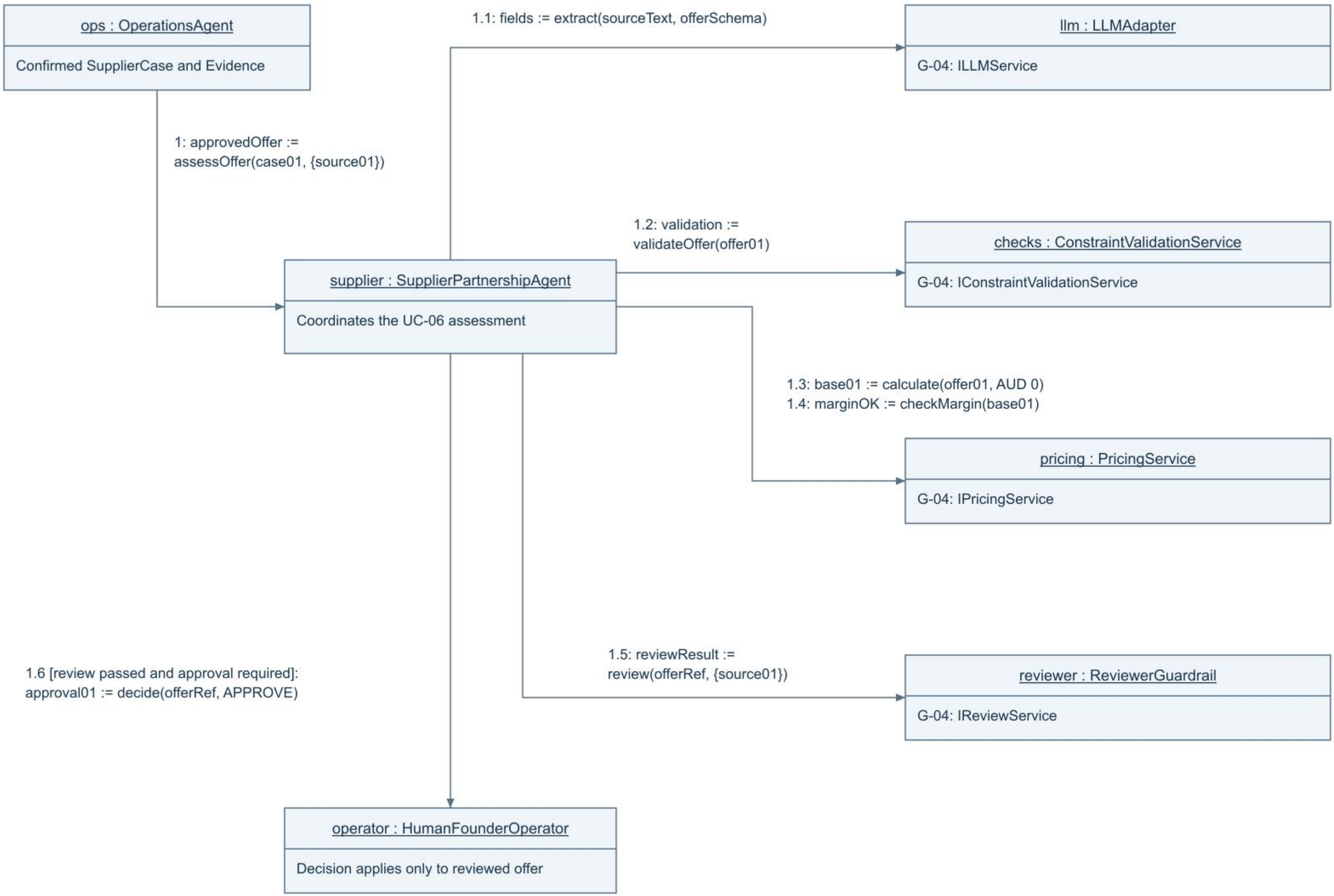
Snapshot: 10 Oct 2026, 09:00 AEST. One approved transport offer selected in a DRAFT itinerary.
All values are simulated fixtures; this is a partial object view, not a complete feasible itinerary.



Underlined headers denote instances; solid lines denote links. Link names follow G-04 associations.
HumanApproval.subject references offer01; no unmodelled class association is introduced. Minimum-margin fixture: AUD 15.00.
The Paid subscription is planning context; this base offer uses no member discount and sends no commercial advertisement.

G-05B | Supplier assessment collaboration

Prepared by Jason | UC-06 successful restricted-offer scenario | SUP-AH-01, SUP-01, SUP-02



Read numbered messages, not vertical time. Each solid link carries a directed synchronous call; ":=" names its returned value. 1.2 and 1.4 must pass before 1.5; the depicted human decision is APPROVE. Failure and revision paths remain in SD-04 / UC-06. After 1.6, the supplier persists the approved result and trace locally, then returns message 1. Operations retains external delivery. Object and operation names match current G-04. extractFields / validateAndPrice in SD-04 are higher-level responsibility labels, refined here into extract, validateOffer, calculate and checkMargin. This view adds no advertising or payment integration.

G-05C | Supplier assessment internal structure

Prepared by Jason | UC-06 | SUP-AH-01, SUP-01, SUP-02 | Runtime refinement of current G-04

ops : OperationsAgent

External to this collaboration

operator : HumanFounderOperator

External to this collaboration

SupplierAssessmentContext

Structured classifier for this runtime collaboration; internal roles are existing G-04 types.

intake port: provides assessOffer
Delegation to supplier.assessOffer

supplier : SupplierPartnershipAgent [1]

assessOffer(case, evidence)
Coordinates checks, review and approval

decision port: requires decide
Delegates requests to the human operator

service port
provides ILLMService

service port
provides IConstraintValidationService

service port
provides IPricingService

service port
provides IReviewService

llm : LLMAdapter [1]

extract / explain

checks : ConstraintValidationService [1]

validateOffer

pricing : PricingService [1]

calculate / checkMargin

reviewer : ReviewerGuardrail [1]

review

supplier requires
ILLMService

supplier requires
IConstraintValidationService

supplier requires
IPricingService

supplier requires
IReviewService

Small squares are ports. Plain lines are connectors, not execution order. Service connectors match required/provided contracts.
Part labels are not underlined: they denote typed roles with multiplicity, not the concrete objects shown in G-05A.

Scope: supplier assessment collaboration inside TripMate AI. Operations and the human are outside this context, not outside all software.
SupplierAssessmentContext and the shown ports are a runtime design refinement; they do not add new business-domain classes.
Persistence uses the existing AgentRun / TraceEntry records and local application storage; repositories and an approval queue are not expanded into additional services here. Waiting for a human is a stored status, not a new message broker.
The full group runtime model must integrate planning, exception handling and the shared UI/storage around this contribution.